

On a variant of the 12th Delfino problem

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Abstract

Assuming that M_n , the canonical inner model with n -many Woodin cardinals exists, we force a model where every Σ^1_{n+2} set is Lebesgue measurable, has the Baire property and the Σ^1_{n+3+m} -uniformization property holds for every $m \in \omega$. Additionally this universe has a Δ^1_{n+3} -definable wellorder of the reals which answers a question from S. D. Friedman and R. Schindler from 1999.

1 Introduction

The structural theory of the projective hierarchy exhibits a profound dichotomy between regularity properties—typically induced by large cardinals and determinacy hypotheses—and the existence of definable wellorders, which are traditionally derived from the fine structure of inner models. Under full Projective Determinacy (PD), the projective sets exhibit complete regularity, precluding the existence of any projective wellorder of the reals. Conversely, in canonical inner models, definability and regularity are sharply demarcated.

The primary motivation for this article is rooted in a version of Woodin’s 12th Delfino problem, which in its original formulation asks whether PD is strictly equivalent to all its structural consequences for the projective hierarchy.

A localized formulation of this question, first posed by S. Friedman and R. Schindler (see [4] and [21]) goes as follows: if we observe a behavior of the projective sets of reals which is the same as the one in some M_{n+1} , the canonical inner model with $n + 1$ Woodin cardinals, is it true that the local

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hypothesis of projective determinacy, namely Π_{n+1}^1 -determinacy or even the weaker (if $n+1$ is odd) Δ_{n+1}^1 -determinacy, must be true? Specifically:

Question 1. *Assume $n \geq 3$. Suppose that every Σ_{n+2}^1 set is Lebesgue measurable and has the Baire property, and there is a Δ_{n+3}^1 -definable wellorder of the reals. Does Δ_{n+1}^1 -determinacy hold?*

The main result of this work demonstrates that the answer to this question is no, even if we additionally add another feature of M_{n+1} , namely the Σ_{n+3+m}^1 -uniformization property for every $m \in \omega$.

Theorem 1.1. *Assume that M_n , the canonical inner model with n Woodin cardinals, exists. Then there is a generic extension of M_n in which Martin's Axiom ($\mathbf{MA}_{\omega_1}$) holds, every Σ_{n+2}^1 set is Lebesgue measurable and has the Baire property, Σ_{n+2+m}^1 -uniformization holds for all $m \in \omega$, and the universe admits a Δ_{n+3}^1 -definable wellorder of the reals.*

This theorem should be contextualized against the properties of the projective sets of reals in M_{n+1} . In M_{n+1} , the presence of $n+1$ Woodin cardinals yields Π_{n+1}^1 -determinacy by the Martin-Steel theorem ([16]), guaranteeing that all sets in Σ_{n+2}^1 are Lebesgue measurable and possess the Baire property via unfolding games. Simultaneously, the fine structure of M_{n+1} provides a good Σ_{n+3}^1 -definable wellorder of the reals (see [24]), which in turn implies Σ_{n+3+m}^1 -uniformization for every $m \in \omega$ by an old result of J. Addison (see [2]). Thus, the universe we force over M_n mimics the behaviour of the reals in M_{n+1} , but uses significantly less large cardinal strength. As in M_n , Δ_{n+1}^1 -determinacy fails ¹ the negative answer to the Friedman-Schindler question follows.

Our proof takes advantage of a result of G. Hjorth ([7], Corollary 2.4) who observed that in the presence of “every real has a sharp”, $\mathbf{MA}_{\omega_1}$ implies that every Σ_3^1 -set is Lebesgue measurable and has the Baire property. Hjorth's result relies on the classical Martin-Solovay tree ([15]) denoted by T_2 . Using the generalized versions T_n of the Martin-Solovay tree, Hjorth's result can be pushed up to higher projective pointclasses thus establishing a path for a proof of the theorem: working over M_n as the ground model, the objective is to force $\mathbf{MA}_{\omega_1}$ while simultaneously introducing a Δ_{n+3}^1 -definable wellorder and preserving global tail Σ -uniformization. Then apply a generalization

¹The failure of Δ_{n+1}^1 -determinacy in M_n can be seen via comparison.

Woodin's inner model characterization establishes that Δ_{n+1}^1 -determinacy is equivalent to the statement that for every real x , there exists a countable, iterable x -mouse with n Woodin cardinals.

If Δ_{n+1}^1 -determinacy were to hold in M_n , then M_n would contain a set sized, iterable mouse N with n Woodin cardinals (see [20]). Coiterating N with M_n yields a proper initial segment of an iterate of M_n containing n Woodin cardinals. However, by the minimality of M_n , no proper initial segment of M_n (or its iterates) can contain n Woodin cardinals. This contradiction demonstrates that no such mouse N exists in M_n , and thus Δ_{n+1}^1 -determinacy must fail in M_n .

of Hjorth’s argument to argue for Lebesgue measurability and the Baire property. To achieve the former, we adapt the coding techniques pioneered in [11], which possess the necessary flexibility to accommodate the forcing of $\mathbf{MA}$.

There are several technical obstacles in the suggested path of our proof. We list two of them. First there is considerable tension between forcing a projectively definable wellorder of the reals and the Σ_{n+3+m}^1 -uniformization on the one hand and forcing $\mathbf{MA}_{\omega_1}$ on the other. Indeed forcing for the first objective is in need of a very tight control over certain projective predicates, whereas forcing for $\mathbf{MA}_{\omega_1}$ necessarily creates a substantial amount of generically added chaos. A second substantial technical obstacle arises because the methods in [11] inherently rely on L as the ground model, which behaves considerably nicer when applying coding forcings than M_n , in particular in iterations of length ω_2 which we must use. To bridge this gap, we implement and refine the forcing machinery introduced in [9] and [12]. The main innovation here is to replace M_n with an intermediate generic extension that exhibits greater robustness under the subsequent coding forcings required for the definable wellorder and the uniformization. Notably, this intermediate extension is itself constructed via coding forcings, meaning the entire architecture functions as a “coding over codes” construction, effectively circumventing the shortcomings of M_n .

We end with a brief description of the organization of this article. In the first sections we introduce the tools we shall use in our proof. This includes several properties of M_1 and M_n , and three coding forcings which we will weave together for a suitable coding machinery. The main proof will be shown with M_1 as the ground model and we later indicate how to implement the changes one has to make in order to lift the proof to M_n . We note that working over M_1 is not necessary, the proof would also work when applied to the universe $L^\#$, the least inner model where every set has a sharp. But using M_1 instead makes the lifting arguments to arbitrary M_n more straightforward.

2 Canonical inner models with Woodin cardinals and the trees T_n

2.1 The canonical inner model with one Woodin cardinal

We will first introduce some of the properties of M_1 which are crucial for our needs. We explicitly single out M_1 as we will prove the theorem in full detail only with M_1 as our ground model and will later indicate how to alter the proof in order to get the full result. Recall that M_1 is a proper class premouse containing a Woodin cardinal (see [24], pp. 81 for a definition of M_1). Every initial segment $\mathcal{J}_\beta^{M_1}$ is ω -sound and 1-small, where we say that

a premouse $\mathcal{M}$ is 1-small iff whenever κ is the critical point of an extender on the $\mathcal{M}$ -sequence then

$$\mathcal{J}_\kappa^\mathcal{M} \models \neg \exists \delta (\delta \text{ is Woodin}).$$

The reals of M_1 admit a Σ_3^1 -definable good wellorder (see [24], Theorem 4.5), the definition of the good wellorder makes crucial use of a weakened notion of iterability, the so-called Π_2^1 -iterability which we shall introduce.

Let $\mathcal{M}$ be a premouse, $\mathcal{T}$ be an ω -maximal iteration tree b a branch through $\mathcal{T}$ and α an ordinal. Then b is α -good if, whenever $\mathcal{N} = \mathcal{M}_b^\mathcal{T}$ or $\mathcal{N}$ is the α -th iterate of some initial segment $\mathcal{P} \trianglelefteq \mathcal{M}_b^\mathcal{T}$ using a single extender E (and its images under the iteration map) on the $\mathcal{P}$ -sequence, then α is in the wellfounded part of $\mathcal{N}$. Then we say that a premouse $\mathcal{M}$ is Π_2^1 -iterable, if player II has a winning strategy in the game $\mathcal{G}'_\omega(\mathcal{M}, 1)$, where $\mathcal{G}'_\omega(\mathcal{M}, 1)$, is defined just as the ordinary weak two player game $W\mathcal{G}_\omega(\mathcal{M}, 1)$ (see e.g. [23] pp. 65 for a definition), with the exception that player I not only plays an ω -maximal, countable putative iteration tree $\mathcal{T}$ but additionally has to play a countable ordinal $\alpha < \aleph_1^{M_1}$. Then player II does not have to play a wellfounded branch through $\mathcal{T}$ (as it would be the case for iterability), but instead can play a cofinal branch b through $\mathcal{T}$ such that b is α -good in order to win.

The winning strategy for II for $\mathcal{G}'_\omega(\mathcal{M}, 1)$ guarantees that $\mathcal{M}$ can be compared to any countable premouse which is an initial segment of M_1 .

We briefly introduce a couple of fundamental inner model theoretic notions which will play a crucial role in the arguments to come.

Definition 2.1. *Let $\mathcal{T}$ be a k -maximal iteration tree of limit length on some premouse $\mathcal{M}$. Then we let*

$$\delta(\mathcal{T}) = \sup\{E_\alpha^\mathcal{T} : \alpha < lh(\mathcal{T})\}$$

Moreover we let

$$\begin{aligned} \mathcal{M}(\mathcal{T}) &:= \text{unique passive } \mathcal{P} \text{ such that } Ord^\mathcal{P} = \delta(\mathcal{T}) \text{ and} \\ &\forall \alpha < \delta(\mathcal{T}) (\mathcal{M}(\mathcal{T}) \text{ agrees with } \mathcal{M}_\alpha^\mathcal{T} \text{ below } lh(E_\alpha^\mathcal{T})). \end{aligned}$$

We also need the so-called zipper lemma (see [23], Theorem 6.10).

Theorem 2.2. *Let c and c' be distinct cofinal branches of the $k(\leq \omega)$ -maximal iteration tree $\mathcal{T}$ on the premouse $\mathcal{M}$. Let $A \subset \delta(\mathcal{T})$ and assume that $\delta(\mathcal{T}), A \in wfp(\mathcal{M}_c^\mathcal{T}) \cap wfp(\mathcal{M}_{c'}^\mathcal{T})$. Then*

$$\mathcal{M}_c^\mathcal{T} \models \exists \kappa < \delta(\mathcal{T}) (\kappa \text{ is } A\text{-strong up to } \delta(\mathcal{T}))$$

Last we recall what $\mathcal{Q}$ -structures are.

Definition 2.3. Let $\mathcal{M}$ be a premouse, $\mathcal{T}$ be a $k(\leq \omega)$ -maximal iteration tree on $\mathcal{M}$, let b be a cofinal, wellfounded branch through $\mathcal{T}$ and let γ be the least ordinal, if there is one, such that either

$$\omega\gamma < \text{On}^{\mathcal{M}_b^{\mathcal{T}}} \text{ and } \mathcal{J}_{\gamma+1}^{\mathcal{M}_b^{\mathcal{T}}} \models \delta(\mathcal{T}) \text{ is not Woodin,}$$

or

$$\omega\gamma = \text{On}^{\mathcal{M}_b^{\mathcal{T}}} \text{ and } \rho_{n+1}(\mathcal{J}_\gamma^{\mathcal{M}_b^{\mathcal{T}}}) < \delta(\mathcal{T})$$

for some $n < \omega$ such that $n+1 \leq k$ if $D^{\mathcal{T}} \cap b = \emptyset$. We set

$$\mathcal{Q}(b, \mathcal{T}) := \mathcal{J}_\gamma^{\mathcal{M}_b}$$

if there is such a γ , and let $\mathcal{Q}(b, \mathcal{T})$ be undefined else.

The next lemma is folklore, but we could not find a proof in the literature, this is why we include it here².

Lemma 2.4. Let $\mathcal{M}$ and $\mathcal{N}$ be ω -sound premice which both project to ω . Assume that $\mathcal{M}$ is an initial segment of M_1 and $\mathcal{N}$ is Π_2^1 -iterable, and let Σ denote the winning strategy for player II in $\mathcal{G}_\omega(\mathcal{M}, \omega_1 + 1)$. Then we can successfully compare $\mathcal{M}$ and $\mathcal{N}$ and consequentially $\mathcal{M} \triangleleft \mathcal{N}$ or $\mathcal{N} \triangleleft \mathcal{M}$.

Proof. We compare $\mathcal{M}$ and $\mathcal{N}$ via iterating away the least disagreement, producing iteration trees $\mathcal{T}$ on the $\mathcal{M}$ -side and $\mathcal{U}$ on the $\mathcal{N}$ -side. We shall show that at every limit stage $\beta \leq \omega_1$, there is a cofinal and wellfounded branch for each side and then use the usual finestructural argument to finish the proof. At limit stages, if $\mathcal{T}$ is the tree we have created so far on the $\mathcal{M}$ -side, we pick the branch b and $\mathcal{M}_b^{\mathcal{T}}$ according to the winning strategy Σ . We shall describe the strategy Γ for player II on $\mathcal{N}$ which ensures a successful comparison of $\mathcal{M}$ and $\mathcal{N}$. For that it is sufficient to determine which branch c player II should pick at limit length trees $\mathcal{U}$ such that the resulting model $\mathcal{M}_c^{\mathcal{U}}$ is wellfounded.

Assume that we arrive at a countable limit stage $\gamma = lh(\mathcal{T}) = lh(\mathcal{U})$, where $\mathcal{T}$ and $\mathcal{U}$ denote the trees we have created so far and assume first that $\delta(\mathcal{T}) = \delta(\mathcal{U})$, where $\delta(\mathcal{T}) = \sup\{lh(E_\alpha^{\mathcal{T}}) : \alpha < \gamma\}$. As $\mathcal{M} \triangleleft M_1$, and the latter is 1-small, we know that $\mathcal{M}_b^{\mathcal{T}}$ is 1-small as well. Consequentially, there is a $\mathcal{Q}$ -structure $\mathcal{Q}(b, \mathcal{T}) \triangleleft \mathcal{M}_b^{\mathcal{T}}$, which has to be of the form $L_\alpha(\mathcal{M}(\mathcal{T}))$, for some $\alpha < \omega_1$, as by 1-smallness you can not have another extender on the sequence until you destroy the Woodin cardinal. Let α be the ordinal height of that $\mathcal{Q}$ -structure. Note that there is failure of $\delta(\mathcal{T})$ being Woodin which is definable over $L_\alpha(\mathcal{M}(\mathcal{T}))$ by definition.

We shall now define which branch c player II shall pick in $\mathcal{U}$. As $\mathcal{N}$ is Π_2^1 -iterable, we know that there is a branch c through $\mathcal{U}$, such that α is in

²We thank F. Schlutzenberg for several discussions on the topic which were very valuable.

the wellfounded part of $\mathcal{M}_c^{\mathcal{U}}$ (or in a linear iteration of one $\mathcal{M}_c^{\mathcal{U}}$'s extenders, but we suppress this case as it will eventually result in exactly the same argument using the elementary embedding of the linear, iterated ultrapower). As $\mathcal{T}, \mathcal{U}$ are the result of iterating away the least disagreement and $\delta(\mathcal{T}) = \delta(\mathcal{U})$, we obtain that $\mathcal{M}_b^{\mathcal{T}}|\delta(\mathcal{T}) = \mathcal{M}_c^{\mathcal{U}}|\delta(\mathcal{U}) = \mathcal{M}(\mathcal{T}) = \mathcal{M}(\mathcal{U})$, and the $\mathcal{Q}(b, \mathcal{T})$ -structure $L_\alpha(\mathcal{M}(\mathcal{T}))$ must also be contained in the wellfounded part of $\mathcal{M}_c^{\mathcal{U}}$. Hence there is a witness to $\delta(\mathcal{T})$ not being Woodin which is definable over $L_\alpha(\mathcal{M}(\mathcal{T}))$, so the $\mathcal{Q}(c, \mathcal{U})$ -structure must equal $L_\alpha(\mathcal{M}(\mathcal{T}))$ as long as $\mathcal{M}_c^{\mathcal{U}} \cap \text{Ord} > \alpha$ (if $\mathcal{M}_c^{\mathcal{U}} \cap \text{Ord} = \alpha$, we are done as c will be wellfounded). This argument is uniform for the branch c we pick so that $L_\alpha(\mathcal{M}(\mathcal{T}))$ is also the $\mathcal{Q}(c', \mathcal{U})$ -structure for any other sufficiently wellfounded branch c' through $\mathcal{U}$.

We claim that $\mathcal{M}_c^{\mathcal{U}}$ is fully wellfounded. Indeed if we let $\alpha' > \alpha$ be countable, then there must be a branch c' through $\mathcal{U}$ such that c' is α' -good. But now $c' = c$, as otherwise the two distinct branches c and c' through $\mathcal{U}$ will witness that $\delta(\mathcal{T}) = \delta(\mathcal{U})$ is A -strong in $\mathcal{M}_c^{\mathcal{U}}$ up to $\delta(\mathcal{T})$ for every $A \subset \delta(\mathcal{T})$, $A \in \text{wfp}(\mathcal{M}_c^{\mathcal{U}}) \cap \text{wfp}(\mathcal{M}_c^{\mathcal{T}})$. But this contradicts the fact that there is a witness definable over $L_\alpha(\mathcal{M}_b^{\mathcal{T}})$ to the fact that $\delta(\mathcal{T})$ is not Woodin. This ends the discussion of the case where $\delta(\mathcal{T}) = \delta(\mathcal{U})$ being countable.

We shall now consider the second case, where one side stops to use extenders at some point in the comparison process. If it is the $\mathcal{N}$ -side, which stops playing extenders at some point then there is nothing to show. If it is the $\mathcal{M}$ -side who stops playing extenders, then there is a last model $\mathcal{M}_\beta$, while on the $\mathcal{N}$ -side, $\mathcal{N}(\mathcal{U})$ exists and is contained in $\mathcal{M}_\beta$. But then there is a model of the form $L_\alpha(\mathcal{M}_\beta|\delta(\mathcal{U}))$ which codes a failure of $\delta(\mathcal{U})$ being Woodin, and the model can be used on the $\mathcal{N}$ -side just as above to find the unique, wellfounded branch through $\mathcal{U}$.

What remains to show is that player II can find a wellfounded branch through $\mathcal{U}$ after ω_1 many stages. Assume that this is not the case, then after ω_1 -many rounds of the comparison game, there is a putative, ω -maximal iteration tree $\mathcal{U}$ on $\mathcal{N}$ with no wellfounded branch. We force with the Lévy collapse $\text{Col}(\omega, \omega_1)$, and let g be an $(M_1, \text{Col}(\omega, \omega_1))$ -generic filter. As $\mathcal{N} \in M_1$ is Π_2^1 -iterable, which is a Π_2^1 -notion, we know that $\mathcal{N}$ is Π_2^1 -iterable in $M_1[g]$ as well. Thus, in $M_1[g]$, there is a cofinal wellfounded branch for $\mathcal{U}$ and it has to be the unique one, by the argument above. Consequentially the branch b is ordinal definable in $M_1[g]$ using $\mathcal{N}, \mathcal{M}, \mathcal{U}, \mathcal{T}$ as parameters, and by the homogeneity of the Lévy collapse, $b \in M_1$ as well, which is what we wanted to show.

To finish the proof, we have to show that $\mathcal{N} \leq \mathcal{M}$ or vice versa. This is just a repetition of the finestructural fact that sufficiently iterable, ω -sound and ω -projecting premice line up (see [23], Corollary 3.12).

As both premice must be countable, the above argument, together with the proof of the comparison lemma (see [23], Theorem 3.11) shows that we can successfully compare them in a $< \omega_1$ -long process, while ensuring that

the trees $\mathcal{T}$ on $\mathcal{M}$ and $\mathcal{U}$ on $\mathcal{N}$ have last models $\mathcal{M}_\alpha$ and $\mathcal{N}_\beta$ such that one is an initial segment of the other and such that the branch leading up to the shorter one does not drop in model or degree. Let us assume without loss of generality, that $\mathcal{M}_\alpha \leq \mathcal{N}_\beta$ and $[0, \alpha]$ does not drop in model or degree. Then, as $\mathcal{M}$ projects to ω , there are no extenders over $\mathcal{M}$ with critical point $< \rho_\omega(\mathcal{M})$, hence if α would be greater than 0, there would be a drop in model or degree in $[0, \alpha]$ which is a contradiction. So $\alpha = 0$, and we turn to β . If $\beta \neq 0$, then, as $\rho_\omega(\mathcal{N}) = \omega$, $\mathcal{N}_\eta$ is not ω -sound, and consequentially $\mathcal{N}_\beta$ is not ω -sound, hence $\mathcal{M}_\alpha$ is a proper initial segment of $\mathcal{N}_\beta$ and is countable there, as $\rho_\omega(\mathcal{M}) = \omega$ and by soundness. But then the iteration from $\mathcal{N}$ leading up to $\mathcal{N}_\beta$ will not move $\mathcal{M}$, and $\mathcal{M}$ is an initial segment of $\mathcal{N}$, which is what we wanted. $\square$

It is relatively straightforward to check that the set of reals which code Π_2^1 -iterable, countable premice is itself a Π_2^1 -definable set in the codes (see [24], Lemma 1.7). Modulo the last lemma, this implies that there is a nice definition of a cofinal set of countable initial segments of M_1 in ω_1 -preserving forcing extensions $M_1[G]$ of M_1 , (in fact this definition holds in all outer models of M_1 with the same ω_1):

Lemma 2.5. *Let $M_1[G]$ be an ω_1 -preserving forcing extension of M_1 . Then in $M_1[G]$ there is Π_2^1 -definable set $\mathcal{I}$ of premice which are of the form $\mathcal{J}_\eta^{M_1}$ for some $\eta < \omega_1$. $\mathcal{I}$ is defined as*

$$\mathcal{I} := \{\mathcal{M} \text{ ctbl premouse} : \mathcal{M} \text{ is } \Pi_2^1\text{-iterable, } \omega\text{-sound and projects to } \omega\},$$

and the set

$$\{\eta < \omega_1 : \exists \mathcal{N} \in \mathcal{I} (\mathcal{N} = \mathcal{J}_\eta^{M_1})\}$$

is cofinal in ω_1 .

In particular $M_1|_{\omega_1} = \mathcal{J}_{\omega_1}^{M_1}$ is definable as the union of the elements in $\mathcal{I}$ in ω_1 -preserving generic extensions of M_1 . We write

$$N = M_1|_{\omega_1} = \mathcal{J}_{\omega_1}^{M_1}$$

for the formula

$$N = \bigcup \{\mathcal{M} \mid \mathcal{M} \in \mathcal{I}\}.$$

Note that the formula $N = M_1|_{\omega_1}$ is not absolute, so if some model satisfies it, it is not guaranteed that N really is $M_1|_{\omega_1}$ or even some countable initial segment of M_1 . We shall need later that $M_1|_{\omega_1}$ can define a $\diamond$ -sequence in the same way as L_{ω_1} can. Indeed, as M_1 has a Δ_3^1 -definable wellorder of the reals whose definition relativizes to $M_1|_{\omega_1}$ we can repeat Jensen's original proof in M_1 to construct a candidate for the $\diamond$ -sequence, via picking at every

limit stage $\alpha < \omega_1$ the $<_{M_1}$ -least pair $(a_\alpha, c_\alpha) \in P(\alpha) \times P(\alpha)$ which witnesses that the sequence we have created so far is not a $\diamond$ -sequence. The proof that this defines already a witness for $\diamond$ is finished as usual with a condensation argument. Hence we shall show that if $\mathcal{J}_\beta^{M_1}$ is least such that $(a_\alpha : \alpha < \omega_1)$ and $(A, C) \in \mathcal{J}_\beta^{M_1}$, where (A, C) is the $<_{M_1}$ -least witness for $(a_\alpha)_{\alpha < \omega_1}$ not being a $\diamond$ -sequence, then there is a countable $N < \mathcal{J}_\beta^{M_1}$ such that the transitive collapse $\bar{N}$ is an initial segment of M_1 .

To see that in fact every such N collapses to an initial of M_1 , recall the condensation result as in [23], Theorem 5.1, which we can state in our situation as follows:

Theorem 2.6. *Let $\mathcal{M}$ be an initial segment of M_1 . Suppose that $\pi : \bar{N} \rightarrow \mathcal{M}$ is the inverse of the transitive collapse and $\text{crit}(\pi) = \rho_\omega^{\bar{N}}$, then either*

1. $\bar{N}$ is a proper initial segment of $\mathcal{M}$, or
2. there is an extender E on the $\mathcal{M}$ -sequence such that $\text{lh}(E) = \rho_\omega^{\bar{N}}$, and $\bar{N}$ is a proper initial segment of $\text{Ult}_0(\mathcal{M}, E)$.

We shall argue, that in our situation, the second case is ruled out, hence every $N < \mathcal{J}_\beta^{M_1}$ collapses to an initial segment of M_1 . Indeed, due to the ω -soundness of $\mathcal{J}_\beta^{M_1}$, every $N < \mathcal{J}_\beta^{M_1}$ will satisfy that

$$\rho_\omega^N = \rho_\omega^{\mathcal{J}_\beta^{M_1}} = \omega_1^{\mathcal{J}_\beta^{M_1}},$$

hence $\text{crit}(\pi) = \omega_1^{\bar{N}} = \rho_\omega^{\bar{N}}$ by elementarity of π .

But $\bar{N}|\omega_1^{\bar{N}} = N|\omega_1^N$, and as $\bar{N}|\omega_1^{\bar{N}}$ thinks that ω is its largest cardinal, $N|\omega_1^N$ must believe this as well. But then there can not be an extender on the N -sequence which is indexed at $\omega_1^{\bar{N}}$, as otherwise $N|\omega_1^{\bar{N}}$ would think that $\omega_1^{\bar{N}}$ is inaccessible, which is a contradiction. Hence, the condition $\text{lh}(E) = \rho_\omega^{\bar{N}}$ is impossible and all that is left is case 1, so $\bar{N}$ is an initial segment of M_1 .

This shows that Jensen's construction of a $\diamond$ -sequence succeeds when applied to M_1 . It is straightforward to verify that the recursive construction can be carried out in $M_1|\omega_1$ by absoluteness. Consequently the $\diamond$ -sequence is a Σ_1 -definable class over $M_1|\omega_1$.

We shall use the $\diamond$ -sequence to construct an ω -length sequence of independent Suslin trees $\tilde{T}^0$, but we will postpone this to the point where this sequence becomes necessary for the constructions.

2.2 The canonical inner model with n -many Woodins

M_n is defined as the canonical minimal inner model that satisfies the statement "There are n Woodin cardinals". It is constructed from an initial segment of $\mathfrak{C}^t$, which is a sequence of tame mice built using full background extenders from the universe. Depending on the smallness of these mice, M_n

is formally obtained either as the limit $\mathcal{N}_\infty$ (if all $\mathcal{N}_\xi$ are n -small) or as the proper class-sized hull of an iterate of a specific active mouse $\mathcal{N}_\xi$. We quickly list the key properties of M_n we shall need in this article. They are all proved in J. Steel's [23].

- Provided that there are n Woodin cardinals in the universe, M_n itself satisfies "There are n Woodin cardinals".
- M_n satisfies that its reals $\mathbb{R}$ possess a Δ_{n+2}^1 well-order.
- The countable initial segments of M_n can be coded by a Π_{n+1}^1 -definable set. This is because the Π_n -iterability condition that characterizes these mice is Π_{n+1}^1 .

The models M_n satisfy a generalization of lemma 2.5.

Lemma 2.7. *Let $M_n[G]$ be an ω_1 -preserving forcing extension of M_n . Then in $M_n[G]$ there is Π_{n+1}^1 -definable set $\mathcal{I}$ of premice which are of the form $\mathcal{J}_\eta^{M_n}$ for some $\eta < \omega_1$. $\mathcal{I}$ is defined as*

$$\mathcal{I} := \{\mathcal{M} \text{ ctbl premouse} : \mathcal{M} \text{ is } \Pi_n^1\text{-iterable, } \omega\text{-sound and projects to } \omega\},$$

and the set

$$\{\eta < \omega_1 : \exists \mathcal{N} \in \mathcal{I} (\mathcal{N} = \mathcal{J}_\eta^{M_n})\}$$

is cofinal in ω_1 .

M_n exhibits a high degree of generic absoluteness in small generic extensions of itself. If M_n is extended to $M_n[g]$ via a forcing poset whose cardinality is less than the least Woodin cardinal in M_n , then $M_n <_{\Sigma_{n+2}^1} M_n[g]$.

This correctness is deeply tied to the absoluteness of higher analogues of the Martin-Solovay trees, denoted by T_i , which project to the universal Σ_{i+1}^1 -sets. These trees are important for our arguments so we introduce them.

2.3 The trees T_n , Weak Homogeneity and Generic Absoluteness

We need the appropriate generalizations of the Martin-Solovay tree T_2 . These trees will be denoted as usual with T_n and will project to a universal Σ_{n+1}^1 set U_{n+1} . These trees T_n will be constructed inside M_{n-1} and we only demand that they project to the local universal Σ_{n+1}^1 -set there, which is sufficient for our needs. As with T_2 , the trees T_n will be absolutely definable in small forcing extensions of M_{n-1} . The key here is that in the presence of Woodin cardinals, the set U_{n+1} is weakly homogeneous Suslin, a notion we quickly re-introduce for the readers convenience.

Let T be a tree on $\omega \times \kappa$, and let $s \in \omega^{<\omega}$. The section of T at s is $T_s = \{u \in \kappa^{|s|} \mid (s, u) \in T\}$. A measure μ on T_s is a countably complete

ultrafilter over T_s . If $s \subseteq t$, a measure ν on T_t projects to a measure μ on T_s if for every $X \subseteq T_s$, $X \in \mu \iff \{u \in T_t \mid u \restriction |s| \in X\} \in \nu$.

Definition 2.8. A tree T on $\omega \times \kappa$ is weakly homogeneous if there exists a system $\bar{\Sigma} = \langle \Sigma_s \mid s \in \omega^{<\omega} \rangle$ such that:

1. For each $s \in \omega^{<\omega}$, Σ_s is a countable set of countably complete ultrafilters on T_s .
2. Coherence: If $s \subseteq t$ and $\nu \in \Sigma_t$, there exists $\mu \in \Sigma_s$ such that ν projects to μ .
3. Countable Completeness: For every real $x \in p[T]$, there exists a sequence of measures $\langle \mu_n \mid n \in \omega \rangle$ such that for each n :
 - $\mu_n \in \Sigma_{x \restriction n}$.
 - μ_{n+1} projects to μ_n .
 - The tower $\langle \mu_n \mid n \in \omega \rangle$ is countably complete. That is, if $A_n \in \mu_n$ for all n , there exists an infinite sequence $f \in \kappa^\omega$ such that $f \restriction n \in A_n$ for all n .

Definition 2.9. A set of reals $A \subseteq \omega^\omega$ is weakly homogeneously Suslin if there exists a weakly homogeneous tree T such that $A = p[T]$.

The utility of weak homogeneity in forcing extensions lies in the preservation of well-foundedness. As a consequence the generalizations T_n of the Martin-Solovay tree will have a similar robustness when defining it in (small) generic extensions.

Theorem 2.10. Let V be a model of set theory. Suppose $T \in V$ is a weakly homogeneous tree via a system of measures $\bar{\Sigma}$. Let $\mathbb{P} \in V$ be a forcing notion such that every measure $\mu \in \Sigma_s$ is κ -complete for one fixed $\kappa > |\mathbb{P}|$. Let $G \subseteq \mathbb{P}$ be a V -generic filter. Suppose that $p[T]$ defines a Σ_n^1 set in V then $p[T]^{V[G]}$ defines the same Σ_n^1 set evaluated in $V[G]$.

2.4 Defining the canonical trees T_n inside M_{n-1} from local determinacy and scales

We fix an integer n , work in M_{n-1} and aim to define tree T_n . Due to the periodicity we need to distinguish the odd and the even case. We shall also argue that these trees are weakly homogeneous.

2.4.1 The odd level

Assume n is odd, so $n = 2m + 1$ for some integer $m \geq 0$. We work inside the canonical inner model M_{n-1} , which contains $n - 1 = 2m$ Woodin cardinals.

Let $P = P_n \subseteq \omega^\omega \times \omega^\omega$ be the universal Π_n^1 set. By Moschovakis' Periodicity Theorem, because n is odd, the pointclass Π_n^1 has the scale property. Thus, P admits a Π_n^1 -scale $\vec{\varphi} = \langle \varphi_i \mid i \in \omega \rangle$.

We define T^* as the tree of this Moschovakis scale. The nodes of T^* consist of finite attempts to build a real alongside the ordinal norms provided by $\vec{\varphi}$. By a (localized version of a) fundamental result of Martin (see [17]), under the assumption of Π_{n-1}^1 -determinacy, the tree of the Moschovakis scale on the universal Π_n^1 set P is a homogeneous tree. Consequently, $p[T^*] = P$, and P is a homogeneously Suslin set.

The universal Σ_{n+1}^1 set, denoted U_{n+1} , is formed by projecting the universal Π_n^1 set P . For any real $a \in \omega^\omega$:

$$a \in U_{n+1} \iff \exists x \in \omega^\omega P(a, x) \quad (1)$$

Since $P(a, x)$ holds if and only if $(a, x) \in p[T^*]$, we define the projected tree T_n by absorbing the search for the real x into the branch space of the tree. For finite sequences $s \in {}^{<\omega}\omega$ with $|s| = k$:

$$T_n = \left\{ \left(s, \langle (t_0, \beta_0), \dots, (t_{k-1}, \beta_{k-1}) \rangle \right) \mid \left(s, t, \langle \beta_0, \dots, \beta_{k-1} \rangle \right) \in T^* \right\} \quad (2)$$

Under this coordinate shift, $p[T_n] = U_{n+1}$.

To establish the regularity properties of U_{n+1} in M_{n-1} , we analyze the homogeneity of T_n . According to Proposition 2.10 in Steel's exposition [25], assuming the existence of a measurable cardinal, a set A is κ -weakly homogeneous if and only if there is a κ -homogeneous set B such that $x \in A \iff \exists y (x, y) \in B$. In our construction, the set P (represented by T^*) is homogeneously Suslin. The set U_{n+1} is exactly the projection $\exists x P(a, x)$. So working in M_{n-1} the projection of the homogeneous tree T^* directly yields that T_n is a weakly homogeneously Suslin tree.

Furthermore, by Corollary 2.21 from [25] following the Martin-Solovay theorem, any weakly homogeneous set is universally Baire. This ensures the desired absolute complementing properties for T_n across generic extensions.

2.4.2 The Even Levels

Let $n = 2k$. We work inside the inner model M_{n-1} , equipped with $n - 1$ Woodin cardinals, to construct the canonical tree T_n projecting to the universal Σ_{n+1}^1 set.

By the Martin-Steel theorem, the $n - 1$ Woodin cardinals in M_{n-1} imply Π_{n-1}^1 -determinacy. By Moschovakis' Periodicity Theorem, this Π_{2k-1}^1 -determinacy entails that the scale property holds for all Π_{n-1}^1 sets inside M_{n-1} . Since the scale property is preserved under existential quantification over the reals ($\exists^\mathbb{R}$), the universal Σ_n^1 set admits a scale within M_{n-1} without requiring further determinacy assumptions.

Let $U \in M_{n-1}$ be the universal Σ_n^1 set. Admitting a scale, U is homogeneously Suslin in M_{n-1} by Martin's theorem (see Theorem 8.5 in [25]). Let S on $\omega \times \kappa$ be the witnessing homogeneous tree, equipped with a coherent system of measures $\bar{\mu} = \langle \mu_s \mid s \in \omega^{<\omega} \rangle$ (cf. Definition 2.2 in [25]), such that $U = p[S]$.

We apply the Martin-Solovay construction (cf. Definition 2.17 in [25]) to S to obtain a tree for the complement of U , the universal Π_n^1 set, denoted P . For any real $x \notin p[S]$, the section tree $S_x = \{u \in {}^{<\omega}\kappa \mid \langle x \restriction |u|, u \rangle \in S\}$ is well-founded and admits a strict rank function $\rho_x : S_x \rightarrow \text{Ord}$. For each $m \in \omega$, the restriction $\rho_{x,m} : S_{x \restriction m} \rightarrow \text{Ord}$ defines an ordinal $[\rho_{x,m}]_{\mu_{x \restriction m}}$ in the ultrapower $\text{Ult}(M_{n-1}, \mu_{x \restriction m})$. The coherence of the measures ensures the canonical embeddings map these ordinals strictly downward:

$$j_{x \restriction l, x \restriction m}([\rho_{x,l}]_{\mu_{x \restriction l}}) > [\rho_{x,m}]_{\mu_{x \restriction m}}$$

for $l < m$.

Let $\theta = \kappa^+$ be the cardinal successor of κ in M_{n-1} . Because the measures μ_s have support κ , any ill-founded tower produces a descending chain bounded by θ . The Martin-Solovay tree $T^* = ms(\bar{\mu}, \theta)$ formally captures the search for such a strictly decreasing sequence across the ultrapowers:

$$T^* = \left\{ \langle s, \langle \alpha_0, \dots, \alpha_{|s|-1} \rangle \rangle \mid s \in {}^{<\omega}\omega \wedge \forall i < |s| \left(\alpha_i \in \text{Ord}^{\text{Ult}(M_{n-1}, \mu_{s \restriction i})} \cap \theta \right) \wedge \forall l < m < |s| (j_{s \restriction l, s \restriction m}(\alpha_l) > \alpha_m) \right\}.$$

A real y is in P if and only if there exists a branch through T^* along y , which certifies the well-foundedness of S_y . Thus, $p[T^*] = P$.

To obtain the universal Σ_{n+1}^1 set U_{n+1} , we project P . Adopting the standard parametrization, for any real a and integer m :

$$\begin{aligned} (a, m) \in U_{n+1} &\iff \exists x \in \omega^\omega P(a, x, m) \\ &\iff \exists x \in \omega^\omega \exists f \in {}^\omega \text{Ord} \forall k \in \omega \left((a \restriction k, x \restriction k, m), f \restriction k \right) \in T^*. \end{aligned}$$

We define the projected tree T_n by absorbing the search for x and f into a single branch space over $\omega \times \text{Ord}$:

$$T_n = \left\{ \left((s, m), \langle (t_0, \beta_0), \dots, (t_{k-1}, \beta_{k-1}) \rangle \right) \mid \left((s, t, m), \langle \beta_0, \dots, \beta_{k-1} \rangle \right) \in T^* \right\}.$$

Consequently, $(a, m) \in U_{n+1} \iff \exists g \in {}^\omega (\omega \times \text{Ord}) \forall k \in \omega ((a \restriction k, m), g \restriction k) \in T_n$. This confirms $U_{n+1} = p[T_n]$.

Weak Homogeneity of the Universal Σ_{n+1}^1 Set

To establish the regularity properties of U_{n+1} in M_{n-1} , we rely on the continuous propagation of homogeneity representations.

First, consider the universal Π_n^1 set P . The set U is homogeneously Suslin via $\bar{\mu}$, which trivially constitutes a weak homogeneity system. To apply Theorem 3.1 of [25], we must identify a Woodin cardinal δ such that $\bar{\mu}$ is δ^+ -complete. Let $\delta = \delta_{n-2}$ be the highest Woodin cardinal in M_{n-1} . By the proof of the Martin-Steel theorem ([16]), there is such a system $\bar{\mu}$ that is δ^+ -complete.

With $\delta = \delta_{n-2}$ and $\bar{\mu}$ being δ^+ -complete, Theorem 3.1 applies directly. The Martin-Solovay tree $T^* = ms(\bar{\mu}, \theta)$ projecting to the complement P is η -homogeneous for each $\eta < \delta_{n-2}$. Because M_{n-1} contains $n - 1$ Woodin cardinals, T^* is a homogeneous tree, making $P = p[T^*]$ a homogeneously Suslin set.

Next, we evaluate the projection. The set U_{n+1} is the real projection $\exists^{\mathbb{R}}$ of P . According to Proposition 2.10 of [25], in the presence of a measurable cardinal, a set is weakly homogeneous if and only if it is the continuous projection of a homogeneous set. Since M_{n-1} contains measurable cardinals and P is homogeneously Suslin, its projection U_{n+1} is weakly homogeneously Suslin. By the Martin-Solovay theorem, this implies U_{n+1} is universally Baire, providing the required absolute complementing trees for generic extensions.

2.5 Summary

Theorem 2.11. *Work in M_{n-1} and define the tree T_n as described above. Then the universal Σ_{n+1}^1 -set U_{n+1} in M_{n-1} is weakly homogeneous.*

The weak homogeneity of T_n guarantees the stability of its descriptive complexity under small forcings.

Theorem 2.12 (Σ_{n+1}^1 -Generic Absoluteness). *Let $\mathbb{P} \in M_{n-1}$ be a small forcing notion, that is $|\mathbb{P}| < \delta_0$, where δ_0 is the least Woodin cardinal in M_{n-1} . If G is $\mathbb{P}$ -generic over M_{n-1} , then:*

$$p[T_n]^{M_{n-1}[G]} = (U_{n+1})^{M_{n-1}[G]}$$

3 The coding forcings which are used

3.1 Coding Reals by Triples of Ordinals

We utilize the coding technique developed by A. Caicedo and B. Veličković (see [3]). Sets coded via their method serve as a suitable ground-model $H(\omega_2)$. Over this $H(\omega_2)$, we apply further coding utilizing independent Suslin trees; in essence, this amounts to coding on top of already coded sets. A key property of the Caicedo-Veličković coding is that ccc forcings cannot introduce new codes. Consequently, the ground-model $H(\omega_2)$ remains definable in ccc generic extensions. This allows us to identify objects from the

ground-model $H(\omega_2)$ —in our case, Suslin trees—and modify them generically by adding cofinal branches or specializing trees via ccc forcings. This ultimately yields a coding mechanism whose codes are easily decoded.

Although several proofs are omitted in the following discussion (details can be found in [9]), it is necessary to describe the Caicedo-Veličković coding in detail, as its features are crucial for the present work.

Definition 3.1. *A $\vec{C}$ -sequence, also known as a ladder system, is a sequence $(C_\alpha : \alpha < \omega_1, \alpha \text{ is a limit ordinal})$ such that for every α , C_α is a cofinal subset of α and has an order type of ω .*

For the rest of this article, we will fix easily definable ladder systems, which are the only ones we will use throughout. Working over M_n , we will use the Δ_{n+2}^1 -definable wellorder $<_n$ of the reals to define $\vec{C} = (C_\alpha \mid \alpha < \omega_1 \cap \text{Lim})$ inductively via picking at each stage $\alpha < \omega_1 \cap \text{Lim}$ the $<_n$ least real coding a ladder $C_\alpha \subset \alpha$. A standard computations yields that this ladder system is $\Sigma_1(\{\omega_1\})$ -definable over $M_n \mid \omega_2$. From now on whenever a ladder system is mentioned when working over some M_n , it is always implied that we talk about this particular ladder system $\vec{C}$.

Given three subsets of natural numbers, $x, y, z \subseteq \omega$, we can define an oscillation function. First, the set x is transformed into an equivalence relation $\sim_x$ defined on $\omega \setminus x$. For natural numbers n, m in the complement of x where $n \leq m$, we define $n \sim_x m$ if and only if the interval $[n, m]$ has no elements in common with x (i.e., $[n, m] \cap x = \emptyset$). This allows us to define the following:

Definition 3.2. *For a triple of subsets of natural numbers (x, y, z) , let $(I_n : n \in k \leq \omega)$ be the sequence of equivalence classes of $\sim_x$ that have a non-empty intersection with both y and z . The oscillation map $o(x, y, z) : k \rightarrow 2$ is then defined as the function satisfying:*

$$o(x, y, z)(n) = \begin{cases} 0 & \text{if } \min(I_n \cap y) \leq \min(I_n \cap z) \\ 1 & \text{otherwise} \end{cases}$$

Next, we will define how appropriate countable subsets of ordinals can be used to code real numbers. For the remainder of this section, we fix a ladder system $\vec{C}$. Suppose $\omega_1 < \beta < \gamma < \delta$ are fixed limit ordinals, each with uncountable cofinality. Let $N \subseteq M$ be countable subsets of δ . Furthermore, assume that $\{\omega_1, \beta, \gamma\} \subseteq N$, and for each $\eta \in \{\omega_1, \beta, \gamma\}$, $M \cap \eta$ is a limit ordinal and $N \cap \eta < M \cap \eta$.

We can use the pair (N, M) to code a finite binary string. Let $\bar{M}$ be the transitive collapse of M , and let $\pi : M \rightarrow \bar{M}$ be the collapsing map. Define $\alpha_M := \pi(\omega_1)$, $\beta_M := \pi(\beta)$, $\gamma_M := \pi(\gamma)$, and $\delta_M := \bar{M}$. These are all countable limit ordinals. Also, set $\alpha_N := \sup(\pi[\omega_1 \cap N])$. The height $n(N, M)$ of α_N in α_M is the natural number defined by:

$$n(N, M) := |\alpha_N \cap C_{\alpha_M}|$$

where C_{α_M} is an element of our pre-fixed ladder system. Since $n(N, M)$ will appear frequently, we will abbreviate it as n . Note that because the order type of each C_α is ω , and $N \cap \omega_1$ is bounded below $M \cap \omega_1$, $n(N, M)$ is indeed a natural number.

Now, we can assign a triple (x, y, z) of finite subsets of natural numbers to the pair (N, M) as follows:

$$x := \{|\pi(\xi) \cap C_{\beta_M}| : \xi \in \beta \cap N\}$$

Note that x is finite because $\beta \cap N$ is bounded in the set C_{β_M} (which is cofinal in β_M and has order type ω). Similarly, we define:

$$y := \{|\pi(\xi) \cap C_{\gamma_M}| : \xi \in \gamma \cap N\}$$

and

$$z := \{|\pi(\xi) \cap C_{\delta_M}| : \xi \in \delta \cap N\}$$

Again, it is straightforward to see that these are finite subsets of natural numbers.

We can then consider the oscillation $o(x \setminus n, y \setminus n, z \setminus n)$ (recalling that $n = n(N, M)$). If the domain of the oscillation function at these points is greater than or equal to n , we write:

$$s_{\beta, \gamma, \delta}(N, M) := \begin{cases} o(x \setminus n, y \setminus n, z \setminus n) \upharpoonright n & \text{if defined} \\ * & \text{otherwise} \end{cases}$$

Here, $*$ denotes an undefined state. Similarly, if $l > n$, we let $s_{\beta, \gamma, \delta}(N, M) \upharpoonright l = *$.

Finally, we can define what it means for a triple of ordinals (β, γ, δ) to code a real number r .

Definition 3.3. *For a triple of limit ordinals $\omega_1 < \beta < \gamma < \delta$, each with uncountable cofinality, we say that it codes a real $r \in 2^\omega$ if there exists a continuous, increasing sequence $(N_\xi : \xi < \omega_1)$ of countable sets of ordinals whose union is δ . This sequence must also satisfy the condition that there is a club $C \subseteq \omega_1$ such that whenever $\xi \in C$ is a limit ordinal, there exists a $\nu < \xi$ for which:*

$$r = \bigcup_{\nu < \eta < \xi} s_{\beta, \gamma, \delta}(N_\eta, N_\xi)$$

We call the sequence $(N_\xi : \xi < \omega_1)$ a reflecting sequence.

Witnesses for this coding can be added using a proper forcing. Conversely, for fixed triples of ordinals, there is a degree of control over the behavior of continuous, increasing sequences on them:

Theorem 3.4 (Caicedo-Veličković). $(\dagger)$ Given ordinals $\omega_1 < \beta < \gamma < \delta < \omega_2$, each with cofinality ω_1 , there exists a proper notion of forcing $\mathbb{P}_{\beta\gamma\delta}$ such that after forcing with it, the following holds: There is an increasing, continuous sequence $(N_\xi : \xi < \omega_1)$ where $N_\xi \in [\delta]^\omega$ and their union is δ . This sequence is such that for every limit ordinal $\xi < \omega_1$ and every $n \in \omega$, there exist $\nu < \xi$ and $s_\xi^n \in 2^n$ satisfying $s_{\beta,\gamma,\delta}(N_\eta, N_\xi) \upharpoonright n = s_\xi^n$ for every η in the interval (ν, ξ) . In this case, we say the triple (β, γ, δ) is stabilized.

$(\ddagger)$ Furthermore, if we fix a real r , there is a proper notion of forcing $\mathbb{P}_r$. This forcing will produce, for a triple of ordinals $(\beta_r, \gamma_r, \delta_r)$ of size and cofinality ω_1 , a reflecting sequence $(P_\xi : \xi < \omega_1)$, where $P_\xi \in [\delta_r]^\omega$ and $\bigcup P_\xi = \delta_r$. Additionally, there is a club $C \subseteq \omega_1$ such that for every limit ordinal $\xi \in C$, there is a $\nu < \xi$ for which:

$$\bigcup_{\nu < \eta < \xi} s_{\beta_r, \gamma_r, \delta_r}(P_\eta, P_\xi) = r.$$

Both partial orders $\mathbb{P}_{\beta\gamma\delta}$ and $\mathbb{P}_r$ which force $(\dagger)$ and $(\ddagger)$ respectively are actually instances of a general class of notions of forcing which were investigated first by J. Moore's in his work on the Set Mapping Reflection Principle (MRP) (see [19]). We shall see later that these forcings never kill Suslin trees. For that reason we have to introduce a couple of notions from [19]. We need first the following local version of stationarity:

Definition 3.5. Let θ be a regular cardinal, X be an uncountable set, let $M < H_\theta$ be a countable elementary submodel which contains $[X]^\omega$ as an element. Then $S \subset [X]^\omega$ is M -stationary if for every club subset C of $[X]^\omega$, $C \in M$ it holds that

$$C \cap S \cap M \neq \emptyset.$$

Definition 3.6. Let X be an uncountable set, $N \in [X]^\omega$ and $x \subset N$ finite. Then the Ellentuck topology on the set $[X]^\omega$ is generated by base sets of the form

$$[x, N] := \{Y \in [X]^\omega : x \subset Y \subset N\}.$$

From now on whenever we say open we mean open with respect to the Ellentuck topology.

Definition 3.7. Let X be an uncountable set, let θ be a large enough regular cardinal so that $[X]^\omega \in H_\theta$. Then a function Σ is said to be open stationary if and only if its domain is a club $C \subset [H_\theta]^\omega$ and for every countable $M \in C$, $\Sigma(M) \subset [X]^\omega$ is open and M -stationary.

Moore has shown that for any open stationary map Σ it is possible to force a reflecting sequence $(N_\xi : \xi < \omega_1)$ with a proper forcing $\mathbb{P}_\Sigma$ (see [19], Theorem 3.1.).

Proposition 3.8 (Moore). *Let Σ be an open stationary function defined on some club $C \subset [H_\theta]^\omega$ with range $P([X]^\omega)$ for some uncountable set X . Then there is a proper notion of forcing $\mathbb{P}_\Sigma$ which adds a continuous sequence of models $(N_\xi : \xi < \omega_1)$ (a reflecting sequence) in $\text{dom}(\Sigma)$ such that for every limit ordinal ξ there is a $\nu < \xi$ such that for every η with $\nu < \eta < \xi$, $N_\eta \cap X \in \Sigma(N_\xi)$.*

The forcing $\mathbb{P}_\Sigma$ is defined as expected: for an open stationary map Σ let $\mathbb{P}_\Sigma$ consist of conditions p which are functions $p : \alpha + 1 \rightarrow \text{dom}(\Sigma)$, α countable, which are continuous and $\in$ -increasing, and which additionally satisfy the MRP-condition on its limit points, namely that for every $0 < \nu < \alpha$, ν a limit ordinal, there is a $\nu_0 < \nu$ such that $p(\xi) \cap X \in \Sigma(p(\nu))$ for every ξ in the interval (ν_0, ν) . The order is by extension.

Now, as already mentioned above, both forcings $\mathbb{P}_{\beta\gamma\delta}$ and $\mathbb{P}_r$ which will produce $(\dagger)$ and $(\ddagger)$ respectively are of the form $\mathbb{P}_\Sigma$ (see [3], Lemma 1, 4 and 5).

Proposition 3.9. *There are two open stationary maps Σ_r and $\Sigma_{\beta\gamma\delta}$ such that $\mathbb{P}_{\beta\gamma\delta} = \mathbb{P}_{\Sigma_{\beta\gamma\delta}}$ and $\mathbb{P}_r = \mathbb{P}_{\Sigma_r}$.*

As a consequence, if we show that $\mathbb{P}_\Sigma$ always preserves Suslin trees for Σ an arbitrary open stationary map, we will have proven that $\mathbb{P}_r$ and $\mathbb{P}_{\beta\gamma\delta}$ preserve Suslin trees. This is indeed the case as we will show now. We need a Lemma of T. Miyamoto ([18]) first.

Lemma 3.10. *Let T be a Suslin tree and assume that $\mathbb{P}$ is a proper poset. Let θ be a sufficiently large cardinal. Then the following are equivalent:*

1. $\Vdash_{\mathbb{P}} T$ is Suslin
2. *if $M < H_\theta$ is countable, $\eta = M \cap \omega_1$, and $\mathbb{P}$ and T are in M , further if $p \in \mathbb{P} \cap M$, then there is a condition $q < p$ such that for every condition $t \in T_\eta$, (q, t) is $(M, \mathbb{P} \times T)$ -generic.*

Proposition 3.11. *Let Σ be an arbitrary open stationary map and let $\mathbb{P}_\Sigma$ be as defined above. Then $\mathbb{P}_\Sigma$ preserves Suslin trees.*

Proof. We fix an arbitrary Suslin tree T . For every countable $M < H_\lambda$, if $\eta = M \cap \omega_1$ and $t \in T_\eta$ then by the Suslinness of T , t is (M, T) -generic. As a consequence we can consider the generic extension $M[t]$ and note that $M[t]$ will not add any new countable sets of ordinals to M .

Let Σ be the open stationary map. Recall that $\mathbb{P}_\Sigma$ was defined to consist of conditions $p : \alpha + 1 \rightarrow \text{dom}(\Sigma)$, $\alpha < \omega_1$, which are $\in$ -increasing and continuous and which additionally satisfy that for every limit ordinal ν , $0 < \nu < \alpha$ there is a $\nu_0 < \nu$ such that $p(\xi) \cap X \in \Sigma(p(\nu))$ for every ξ in the interval (ν_0, ν) . The order is by extension.

We let λ be sufficiently large and pick a countable $M < H_\lambda$ which contains $T, \Sigma, \mathbb{P}_\Sigma$, a condition $p \in \mathbb{P}_\Sigma$, and the structure $H_{|\mathbb{P}_\Sigma|+}$. Letting $M \cap \omega_1 = \eta$, then our goal is to produce a stronger condition $q < p$ such that for every $t \in T_\eta$, (q, t) is an $(M, \mathbb{P}_\Sigma \times T)$ -generic condition.

We list $(t_n : n \in \omega)$, the elements of T_η and build the according generic extensions $M[t_n]$. As M and $M[t_n]$ have the same countable sequences of M -elements, there is no difference when talking about Σ and $\mathbb{P}_\Sigma$ in either M or $M[t_n]$. We list all the dense subsets $(D_0, D_1, \dots)$ of $\mathbb{P}_\Sigma$ which we can find in $\bigcup_{n \in \omega} M[t_n]$ and build by recursion a descending sequence of conditions $(p_i : i \in \omega)$ in M , starting at $p_0 := p$ hitting the corresponding D_{i-1} .

Assume that we have already built conditions up to $i \in \omega$. We consider the collection of models N'_i , where each N'_i is a countable elementary submodel of $H_{|\mathbb{P}_\Sigma|+}$ containing $H_\theta, D_i, \mathbb{P}_\Sigma$ and p_i , and build the club of countable structures $C_i := \{N'_i \cap X : N'_i \text{ as just described}\}$. Note that this club will be in M , hence $M \cap H_\theta \in C_i$. Thus the set $M \cap H_\theta$ will be in the domain of Σ and by the definition of Σ , the set $\Sigma(M \cap H_\theta)$ is $M \cap H_\theta$ -stationary and open. So there is an $N_i = N'_i \cap X \in C_i \cap \Sigma(M \cap H_\theta) \cap M$, and by the definition of the Ellentuck topology, there is a finite subset of N_i called x_i such that $[x_i, N_i] \subset \Sigma(M \cap H_\theta)$. We first extend the condition p_i to $q_i := p_i \cup \{(\zeta_i + 1, \text{hull}^{H_\theta}(p_i(\zeta_i) \cup x_i))\}$, for ζ_i the maximum of the domain of p_i . This condition q_i will also be in N'_i as all its defining parameters are, thus as N'_i also contains D_i we can extend the condition q_i to a $p_{i+1} \in N'_i \cap D_i$. Note that as we are working in N'_i , no matter how we extend q_i , the range of the extended condition intersected with X will always be contained in $N_i = N'_i \cap X$, and as $\Sigma(M \cap H_\theta) \supset [x_i, N_i]$, it will also be contained in $\Sigma(M \cap H_\theta)$. Then if we set $q = p_\omega := \bigcup_{i \in \omega} p_i \cup (\omega, (M \cap H_\theta))$ then this will be a condition in $\mathbb{P}_\Sigma$ as it forms an ϵ -increasing, continuous function from $\omega + 1$ to $\text{dom}(\Sigma)$, for which $\forall n \in \omega (q(n) \cap X \in \Sigma(q(\omega)) = M \cap H_\theta)$ is true. By construction q is below p and $(M, \mathbb{P}_\Sigma \times T)$ -generic, thus the forcing preserves the arbitrary Suslin tree T . □

3.2 Almost Disjoint Coding

The second coding method we will use is the almost disjoint coding forcing, developed by R. Jensen and R. Solovay [14]. We will identify subsets of ω with their characteristic functions and use the term “reals” for elements of 2^ω and subsets of ω interchangeably.

Let $F = \{f_\alpha : \alpha < 2^{\aleph_0}\}$ be a family of almost disjoint subsets of ω ; that is, a family such that if $r, s \in F$, then their intersection $r \cap s$ is finite. Let $X \subset \kappa$ (where $\kappa \leq 2^{\aleph_0}$) be a set of ordinals. There exists a ccc forcing, known as the almost disjoint coding $\mathbb{A}_F(X)$, which adds a new real x . This real x codes the set X relative to the family F in the following manner:

$$\alpha \in X \text{ if and only if } x \cap f_\alpha \text{ is finite.}$$

Definition 3.12. *The almost disjoint coding $\mathbb{A}_F(X)$ relative to an almost disjoint family F consists of conditions (r, R) , where $r \in \omega^{<\omega}$ (a finite sequence of natural numbers) and $R \in F^{<\omega}$ (a finite sequence of elements from F). A condition (s, S) is stronger than (r, R) (denoted $(s, S) < (r, R)$) if and only if:*

1. *r is an initial segment of s ($r \subset s$), and R is an initial segment of S ($R \subset S$).*
2. *If $\alpha \in X$, then $r \cap f_\alpha = s \cap f_\alpha$.*

Another variant, possibly due to L. Harrington (see [6]), codes sets of reals relative to a new real. For the following, we fix some definable bijection between finite sequences of integers and ω . For $b \in \omega^\omega$, let $\bar{b}(n)$ denote the natural number that codes the finite sequence $b \upharpoonright n$ (the initial segment of b of length n). A subset $b \subset \omega$ (viewed as a real) gives rise to a new set $S(b) \subset \omega$ if we consider the set of codes of its initial segments: $\{\bar{b}(n) : n \in \omega\}$.

Definition 3.13. *Suppose $A \subset [\omega]^\omega$ (i.e., A is a set of infinite subsets of ω). The almost disjoint coding forcing for A , denoted $\mathbb{A}(A)$, is defined as follows. Conditions are pairs $(s(0), s(1))$ such that $s(0)$ is a finite set of natural numbers and $s(1)$ is a finite subset of the fixed set of reals A . For two conditions $p, q \in \mathbb{A}(A)$, we say $q < p$ (q is stronger than p) if and only if:*

- $p(0) \subset q(0)$ and $p(1) \subset q(1)$.
- $\forall a \in p(1) (S(a) \cap q(0) \subset p(0))$.

Note that $\mathbb{A}(A)$ has the Knaster property, which implies that products of $\mathbb{A}(A)$ satisfy the ccc. Given a set of reals A in the ground model V , forcing with $\mathbb{A}(A)$ has the following effect: The first coordinates of conditions in the generic filter G will combine to form a real a . This real a codes the set A with the help of a predicate for $\omega^\omega \cap V$. In the generic extension $V[G]$, membership in A is characterized as:

$$x \in A \leftrightarrow x \in V \wedge S(x) \cap a \text{ is finite.}$$

This characterization will play a crucial role later. As previously mentioned, our goal is to work towards a universe whose $H(\omega_2)$ is definable in arbitrary generic extensions obtained using ccc forcings. Consequently, we will have definable access to $H(\omega_2)^V$ (the $H(\omega_2)$ of the ground model V). We then use the just-defined forcing to encode information into a single real, ensuring that this information can be correctly decoded in all subsequent ccc extensions of the universe.

3.3 Independent Suslin trees

Suslin trees are used to construct one of the three coding techniques we will use in the proof. Recall that a set theoretic tree $(T, <)$ is a Suslin tree if it is a normal tree of height ω_1 and no uncountable antichain. All the trees which appear in this paper will be normal, thus whenever we talk about trees it is implicitly assumed that these trees are normal.

Recall that for two trees $(T_0, <_{T_0})$ and $(T_1, <_{T_1})$ their tree-product $T_0 \times T_1$ is defined to be the tree which consists of nodes $\{(t_0, t_1) : t_0 \in T_0 \wedge t_1 \in T_1 \wedge \text{height}(t_0) = \text{height}(t_1)\}$, ordered by $(t_0, t_1) <_{T_0 \times T_1} (s_0, s_1)$ if and only if $t_0 <_{T_0} s_0$ and $t_1 <_{T_1} s_1$. From now on whenever we talk about a product of trees, it is always the tree product which is meant. For our purposes it is necessary to iteratively add sequences of blocks of Suslin trees $(\bar{T}_\alpha : \alpha < \kappa)$ such that $\bar{T}$ is itself an ω -length sequence of Suslin trees whose finite subproducts are Suslin again. One can construct such sequences using Jech's forcing which adds a Suslin tree with countable conditions.

Definition 3.14. Let $\mathbb{P}_J$ be the forcing whose conditions are countable, normal trees ordered by end-extension, i.e. $T_1 < T_2$ if and only if $\exists \alpha < \text{height}(T_1) T_2 = \{t \restriction \alpha : t \in T_1\}$

It is wellknown that $\mathbb{P}_J$ is σ -closed and adds a Suslin tree. In fact more is true, the generically added tree T has the additional property that for any Suslin tree S in the ground model $S \times T$ will be a Suslin tree in $V[G]$.

Lemma 3.15. Let V be a universe and let $S \in V$ be a Suslin tree. If $\mathbb{P}_J$ is Jech's forcing for adding a Suslin tree and if g_T is the generic filter producing the tree T then

$$V[g_T] \models T \times S \text{ is Suslin.}$$

Proof. Let $\dot{T}$ be the $\mathbb{P}_J$ -name for the generic Suslin tree which we consider as a forcing notion. We claim that $\mathbb{P}_J * \dot{T}$ has a dense subset which is σ -closed. As σ -closed forcings will always preserve ground model Suslin trees, this is sufficient. To see why the claim is true consider the following set:

$$\{(p, \check{q}) : p \in \mathbb{P}_J \wedge \text{height}(p) = \alpha + 1 \wedge \check{q} \text{ is a node of } p \text{ of level } \alpha\}.$$

It is easy to check that this set is dense and σ -closed in $\mathbb{P}_J * \dot{T}$. □

A similar observation shows that we can add an ω -sequence of such Suslin trees with a fully supported iteration. Even longer sequences of such trees are possible if we lengthen the iteration but for our needs ω -blocks are sufficient.

Lemma 3.16. Let S be a Suslin tree in V and let $\mathbb{P}$ be a fully supported iteration of length ω of forcings $\mathbb{P}_J$. Then in the generic extension $V[G]$ there is an ω -sequence of Suslin trees $\vec{T} = (T_n : n \in \omega)$ such that for any finite $e \subset \omega$ the tree $S \times \prod_{i \in e} T_i$ will be a Suslin tree in $V[\vec{T}]$.

Proof. Let G be a generic filter for $\mathbb{P}$. First we observe that the fully supported product $\prod_{n \in \omega} \mathbb{P}_J$ followed by the fully supported forcing with the product of the generically added trees T_n has a σ -closed dense subset which is defined coordinate-wise as above, thus ground model Suslin trees are preserved. If we pick a finite $e \subset \omega$, then for an arbitrary Suslin tree $S \in V$, $\prod_{i \in e} T_i \times S$ is a Suslin tree in the intermediate model generated by the trees T_i , $i \in e$ over V . This is preserved when passing to the generic extension $V[G]$. $\square$

These sequences of Suslin trees will become important later in our proof, thus they will get a name.

Definition 3.17. Let $\vec{T} = (T_\alpha : \alpha < \kappa)$ be a sequence of Suslin trees. We say that the sequence is an independent family of Suslin trees if for every finite set $e = \{e_0, e_1, \dots, e_n\} \subset \kappa$ the product $T_{e_0} \times T_{e_1} \times \dots \times T_{e_n}$ is a Suslin tree again.

We will use the following preservation result due to T. Miyamoto (see [18]) and independently due to U. Abraham and S. Shelah ([1]).

Theorem 3.18. Let $\mathbb{P} = ((\mathbb{P}_\beta, \dot{\mathbb{Q}}_\beta) : \beta < \delta)$ be a countable support iteration of proper forcings, let S be Suslin tree and assume that for every $\beta < \delta$, $\mathbb{P}_\beta \Vdash \text{“}\dot{\mathbb{Q}}_\beta \text{ preserves } S \text{ as a Suslin tree.”}$ Then S remains a Suslin tree in the generic extension by $\mathbb{P}$.

4 Definition of W_0

We finally have all the ingredients to successfully define the suitable ground model W_0 over which coding arguments will work in a nice way. To form W_0 we start with M_1 as our ground model and let η be an inaccessible limit of inaccessible cardinals in M_1 . Using a η -long iteration we shall produce a generic extension W_0 of M_1 which forces with forcings of the form

1. $\mathbb{P}_{\beta\gamma\delta}$ for triples $\omega_1 < \beta < \gamma < \delta < \omega_2$ of uncountable cofinality.
2. $\mathbb{P}_r$ for reals r .
3. $\mathbb{A}_F(X)$ for $X \subset \omega_1$.
4. $\mathbb{P}_J$ which denotes Jech’s forcing for adding a Suslin tree.

We use countable support for the iteration which has the feature that “Suslin tree preservation” is preserved by theorem 3.18. This preservation is key as we will use the added Suslin trees for a second iteration over W_0 which will use these Suslin trees for coding purposes. The goal is to arrive after η -many steps at a universe W_0 with the following properties

1. $\aleph_1^{M_1} = \aleph_1^{W_0}$.
2. The inaccessible limit of inaccessibles η becomes $\aleph_2$ in W_0 .
3. There is a $\eta = \omega_2$ -length, independent sequence $\vec{T}$ of ω_1 -Suslin trees.
4. For every subset X of ω_1 , there is a real r_X which is an almost disjoint code of X with respect to the canonical M_1 -definable, almost disjoint family of reals $D \subset \omega^\omega \cap M_1$.
5. Every triple of limit ordinals $(\alpha, \beta, \gamma) < \omega_2$ of uncountable cofinality is stabilized in the sense of $(\dagger)$.
6. Every real is coded by a triple of limit ordinals of uncountable cofinality (α, β, γ) .

The actual iteration to obtain W_0 is standard and we are not interested in its specific properties as long as the model W_0 it produces has the 6 properties above. There is a canonical, definable well-order of $P(\omega_1)$ in W_0 .

Definition 4.1. *Let $X, Y \in P(\omega_1)^{W_0}$ then let $X \trianglelefteq Y$ if the antilexicographically least triple of ordinals $(\alpha_0, \beta_0, \gamma_0)$ which code a real r_0 which codes X with the help of the a.d. family F is antilexicographically less or equal than the antilexicographically least triple of ordinals $(\alpha_1, \beta_1, \gamma_1)$ which codes a real r_1 which in turn codes Y .*

The definable wellorder $\trianglelefteq$ of $P(\omega_1)$ unlocks a definition for a canonical sequence of length ω_2 of independent Suslin trees. The first entry of that sequence is, for technical reasons which will become clear later defined differently.

Working over M_1 and using $\diamond$, we define the canonical Suslin tree T on ω_1 . Specifically, let $\langle A_\alpha : \alpha < \omega_1 \rangle$ be a $\diamond$ -sequence. Then T is constructed inductively as follows:

- Initialize the tree with a single root node.
- For every node $x \in T_\alpha$, assign it $\aleph_0$ many immediate successors in $T_{\alpha+1}$.
- To form T_α , we must select branches through the lower part of the tree, $T_{<\alpha}$, to extend.
 - Check if the guessed set A_α happens to be a maximal antichain in $T_{<\alpha}$.
 - If A_α is a maximal antichain, ensure that every branch chosen to be extended into T_α intersects A_α . Because $T_{<\alpha}$ is countable, it is always possible to find enough branches that hit A_α to continue the tree.

- If A_α is not a maximal antichain, extend any set of branches that keeps the level countable.

It is known that the tree T obtained this way is a full Suslin tree (see [5], pp. 431), which means that for every antichain $(a_n \in T \mid n \in \omega)$, the sequence of derived trees $(T_{a_n} \mid n \in \omega)$ is an independent sequence of Suslin trees.

Definition 4.2. *Working in M_1 , we let T be the canonical Suslin tree as defined above and let $(a_n \in T \mid n \in \omega)$ be a list of the nodes of the first level of T . Let T_{a_n} denote the subtree of T with stem a_n and set $T_{a_n} := T_n^0$ for every $n \in \omega$. We let*

$$\vec{T}^0 = \{T_n^0 \mid n \in \omega\} = \{T_{a_n} \mid n \in \omega\}$$

be the resulting independent sequence of Suslin trees.

We start with our fixed independent ω -sequence $\vec{T}^0$ and let $\vec{T}^\alpha$ be the $\trianglelefteq$ -least ω_1 sequence of Suslin trees such that $\bigcup_{\beta < \alpha} \vec{T}^\beta$ concatenated with $\vec{T}^\alpha$ remains an independent sequence of Suslin trees.

As the wellorder $\trianglelefteq$ in fact talks about the reals which are almost disjoint codes for the corresponding elements of $P(\omega_1)$ it will be useful to give that sequence of reals a name as well. For every element X in $P(\omega_1)$, the set of reals which are almost disjoint codes for X is infinite. In the following we nevertheless talk about *the* real r_X which codes $X \in P(\omega_1)$ by which we mean the $\trianglelefteq$ -least such real coding X .

Definition 4.3. *In W_0 , let $(r_i : i < \omega_2)$ be the sequence of reals defined recursively as follows:*

- r_0 is the real which codes a subset of ω_1 , which codes the independent ω -sequence of Suslin trees $\vec{T}^0$.
- r_α , for $\alpha > 0$ is the least real which is an almost disjoint code for the $\trianglelefteq$ -least subset of ω_1 which itself is a code for an ω_1 -sequence of independent Suslin trees $\vec{T}^\alpha$, such that the concatenated sequence of the union of the Suslin trees coded in $(r_i : i < \alpha)$ and $\vec{T}^\alpha$ forms an independent sequence again.

What is very important is that this definable ω_2 -sequence of independent Suslin trees will be definable in certain outer models of W_0 .

Lemma 4.4. *Suppose that W^* is a set-generic, ccc extension of W_0 . Then W^* is still able to define the ω_2 -sequence of independent W_0 -Suslin trees $\vec{T}$.*

Proof. Note first that if $r \in W^*$ is a real coded by a triple of ordinals in (α, β, γ) in W^* , then there is a reflecting sequence $(N_\xi : \xi < \omega_1)$ in W^* , $\bigcup_{\xi < \omega_1} N_\xi = \gamma$, such that for club-many ξ , $r = \bigcup_{\eta \in (\nu, \xi)} s_{\alpha\beta\gamma}(N_\eta, N_\xi)$. As

W^* is a ccc-extension of W_0 , there is a reflecting sequence $(P_\xi : \xi < \omega_1)$ which is an element in W_0 , and such that $C := \{\xi < \omega_1 : P_\xi = N_\xi\}$ is club containing in W^* . Indeed, every element of $(N_\xi : \xi < \omega_1)$ is a countable set of ordinals in W^* , thus can be covered by a countable set of ordinals from W_0 . As a consequence the sequence $(N_\xi : \xi < \omega_1)$ can be transformed into a continuous, increasing sequence $(P_\xi : \xi < \omega_1)$ in W_0 which coincides on its limit point with $(N_\xi : \xi < \omega_1)$, just as desired.

But as ccc extensions preserve stationarity, the set

$$\{\zeta < \omega_1 : \exists \nu < \zeta (\bigcup_{\eta \in (\nu, \zeta))} s_{\alpha\beta\gamma}(P_\eta, P_\zeta) = r)\}$$

which is an element of W_0 must contain a club from W_0 . Hence r is coded by the triple (α, β, γ) already in W_0 .

As a consequence $P(\omega_1)^{W_0}$ is definable in W^* , it will be precisely the set of subsets of ω_1 which have reals which code it with the help of the almost disjoint family F , and such that these reals are themselves coded by triples of ordinals below ω_2 in the sense of $(\ddagger)$.

Thus W^* can define W_0 -Suslin trees and our wellorder $<$ on $P(\omega_1)^{W_0}$, hence will be able to define the ω_2 -sequence of independent Suslin trees of W_0 . □

5 Definition of W_1 and Suitability

We begin to define thoroughly how the forcing over W_0 which will yield W_1 , does look like. We already hinted that, in order to use the independent sequence of Suslin trees $\vec{T}$ we need a new notion for suitable models which will be able to define the sequence of Suslin trees $\vec{T}$ correctly. With the notion of suitability it will become possible to correctly define $\vec{T}$ already in $\aleph_1$ -sized $\mathbf{ZF}^-$ models, as we shall see soon, which is non-trivial, as being an ω_1 -Suslin tree in some $\aleph_1$ -sized $\mathbf{ZF}^-$ model not necessarily implies that this tree is actually Suslin in the real world.

Definition 5.1. *Let M be a transitive model of $\mathbf{ZF}^-$ of size $\aleph_1$. We say that M is pre-suitable if it satisfies the following list of properties:*

1. $M_1|_{\omega_1} \subset M$ so in particular our distinguished ω -sequence of independent Suslin tree $\vec{T}^0$ and our distinguished ladder system $\vec{C}$ are both in M .
2. $\aleph_1$ is the biggest cardinal in M and $M \models \forall x (|x| \leq \aleph_1)$.
3. Every set in M has a real in M which codes it in the sense of almost disjoint coding relative to the fixed family of almost disjoint reals F .

4. Every real in M is coded by a triple of ordinals in M , i.e. if $r \in M$ then there is a triple $(\alpha, \beta, \gamma) \in M$ and a reflecting sequence $(N_\xi : \xi < \omega_1) \in M$ which code r .
5. Every triple of ordinals in M is stabilized in M : for (α, β, γ) there is a reflecting sequence $(P_\xi : \xi < \omega_1) \in M$ which witnesses that (α, β, γ) is stabilized.

Note that the statement “ M is a pre-suitable model” is completely internal in M and hence a $\Sigma_1(\vec{C}, \vec{T}^0)$ and in particular a $\Sigma_1(\{\omega_1\})$ -formula. Further note that by the proof of Lemma 4.4, if W^* is a ccc extension of W_0 and M is a pre-suitable model in W^* then $M \subset W_0$, as ccc extensions will not add new reflecting sequences.

Definition 5.2. Let M be a pre-suitable model. We say that M is W_0 -absolute for Suslinness if $T \in M$ is an element from W_0 and $M \models T$ is Suslin, then T is Suslin in W_0 . Likewise we say that M is W_0 -absolute for stationarity if $S \in M$ and M thinks that S is a stationary subset of ω_1 then S is a stationary subset of ω_1 in W_0 . A pre-suitable model which is W_0 -absolute for stationarity and Suslinness is called suitable.

We have already seen in Lemma 4.4 that ccc extensions of W_0 will still be able to define our ω_2 -sequence of independent W_0 -Suslin trees $\vec{T}$. With the notion of suitability we can localize this property in the following sense:

Lemma 5.3. Let W^* be a ccc extension of W_0 , and let $M \in W^*$ be a suitable model. If M computes the ω_2 -length sequence of independent Suslin trees from W_0 using its local wellorder $\leq_M$, then the computation will be correct, i.e. $\vec{T}^M = \vec{T} \cap M$.

Proof. We shall show inductively that for every $\eta \in M$, the η -th block of $\vec{T}$, $\vec{T}^\eta$ will be computed correctly by M (if an element of M). For $\eta = 0$ this is true as $\vec{T}^0$ is by definition of suitability an element of M . Let $\eta \in M$ and assume by induction that the sequence $\vec{T}$ up to the η -th block is computed correctly by M . We shall show that M computes the $\eta+1$ -th block correctly. Recall that $\vec{T}^{\eta+1}$ was defined to be the $\leq$ -least ω_1 -block of independent Suslin trees such that $\bigcup_{\beta \leq \eta} \vec{T}^\beta$ concatenated with $\vec{T}^{\eta+1}$ remains an independent sequence in W_0 .

Assume for a contradiction that the suitable M computes $\vec{T}'$ as its own different version of $\vec{T}^{\eta+1}$. Thus there is a real r' which codes $\vec{T}'$, and r' itself is coded into a triple of M -ordinals $(\alpha', \beta', \gamma') < \omega_2^M$. Let $r_{\eta+1}$ be the real which codes $\vec{T}^{\eta+1}$. We claim that $r_{\eta+1} < r'$ by which we mean that the least triple of ordinals which codes $r_{\eta+1}$ is antilexicographically less than the least triple which codes r' . Otherwise $r' < r_{\eta+1}$ and by the Suslin-absoluteness of M the independent- M -Suslin trees coded into r' would be an independent ω_1 -sequence of Suslin trees in W_0 , moreover they would still

form an independent sequence when concatenated with $\vec{T}^\eta$ in W_0 which is a contradiction to the way $\vec{T}^{\eta+1}$ was defined.

So $r_{\eta+1} < r'$, so the least triple of ordinals (α, β, γ) coding $r_{\eta+1}$ is antilexicographically less than $(\alpha', \beta', \gamma')$. Note that the suitability of M implies that (α, β, γ) is stabilized in M . Thus there is a reflecting sequence $(P_\xi : \xi < \omega_1)$ in M witnessing this. As W^* is a ccc extension of W_0 we can assume that the sequence is in fact an element of W_0 . At the same time there is a reflecting sequence $(N_\xi : \xi < \omega_1)$ in W_0 which witnesses that $r_{\eta+1}$ is coded by (α, β, γ) . By the continuity of both sequences, there is a club C in W_0 such that $\forall \xi \in C (N_\xi = P_\xi)$. Thus the limit points of C witness that in fact the sequence $(P_\xi : \xi < \omega_1) \in M$ codes $r_{\eta+1}$ as well but the club C is in W_0 , so we need an additional argument to finish. Recall that the suitability of M implies that M is absolute for stationarity, thus if the set $\{\xi < \omega_1 : \exists \nu < \xi (\bigcup_{\zeta \in (\nu, \xi)} s_{\alpha\beta\gamma}(P_\zeta, P_\xi) \neq r_{\eta+1})\}$ would be stationary in M it would be stationary in W_0 which is a contradiction. So M computes $r_{\eta+1}$ correctly and the rest of the inductive argument can be repeated exactly as above to show that $\vec{T}^M = \vec{T} \upharpoonright (M \cap \text{Ord})$ as desired. $\square$

So suitable models will compute $\vec{T}$ correctly, and if we start to write information into $\vec{T}$, a suitable model can be used to read it off. In W_0 cofinally many suitable model below ω_2 exist.

Lemma 5.4. *Work in W_0 . If we set*

$$P := \{\zeta < \omega_2 : \exists M (M \text{ is suitable} \wedge M \cap \text{Ord} = \zeta)\}$$

then P is unbounded in ω_2 .

Proof. Recall the iteration $(\mathbb{P}_\alpha : \alpha < \eta)$ to force W_0 . Whenever we are at an intermediate stage $\kappa < \eta$ such that κ is inaccessible then, if G_κ denotes the generic filter for $\mathbb{P}_\kappa$, $H(\omega_2)^{M_1[G_\kappa]}$ will be a pre-suitable model. Moreover, by the properties of countably supported iterations, Suslin trees in $H(\omega_2)^{M_1[G_\kappa]}$ will remain Suslin trees in W_0 , as the tail iteration $\mathbb{P}_{[\kappa, \eta]}$ is a countable support iteration of proper, Suslin tree preserving notions of forcing over the ground model $M_1[G_\kappa]$. Thus every $H(\omega_2)^{M_1[G_\kappa]}$, for κ inaccessible, is a suitable model which gives the assertion of the Lemma. $\square$

We shall use forcing to obtain a universe in which the just defined set of suitable models

$$U' := \{M : \exists \kappa < \eta (\kappa \text{ is inaccessible} \wedge M = H(\omega_2)^{M_1[G_\kappa]})\}$$

becomes easily definable. This will be our first forcing in order to obtain W_1 over W_0 . We note that every $M \in U'$ is itself coded by a real relative to the almost disjoint family F . We want to use the variant of almost disjoint coding forcing to code up

$U := \{r_M : r_M \text{ is the least almost disjoint code for a subset of } \omega_1 \text{ which codes } M \in U'\}$

into one real r_U . Recall that $\mathbb{A}(U)$ is a forcing of size $|U| = \eta$ which is Knaster and which adds a real r_U such that in $W_0^{\mathbb{A}(U)}$ the following holds:

(*) $\forall x \in 2^\omega \cap W_0 (x \in U \leftrightarrow r_U \cap S(x) \text{ is finite}).$

In a next step we will code the characteristic function of the real r_U into a pattern on $\vec{T}^0$. We use the fact that a Suslin tree can generically be destroyed in two mutually exclusive ways. We can generically add a branch or generically add an antichain without adding a branch. We fix the first ω -block of independent Suslin trees $\vec{T}^0$ and we let $\mathbb{D}(r_U) := \prod_{n \in \omega} \mathbb{P}_n$ with finite support, where

$$\mathbb{P}_n = \begin{cases} T_n^0 & \text{if } r_U(n) = 1 \\ Sp(T_n^0) & \text{if } r_U(n) = 0 \end{cases}$$

and T_n^0 denotes the forcing notion one obtains when forcing with nodes of T_n^0 as conditions, and $Sp(T_n^0)$ denotes Baumgartner's forcing which specializes T_n^0 with finite conditions and which is known to be ccc (see [13]). Iterations of the just described form always have the countable chain condition.

Lemma 5.5. *Let $\vec{T} = (T_\alpha : \alpha < \eta)$ be an independent sequence of Suslin trees of length η . Let $f : \eta \rightarrow 2$ be an arbitrary function and let T_α also denote the partial order when forcing with the tree T_α and let $Sp(T_\alpha)$ be the forcing which specializes the tree T_α .*

Then if we consider the finitely supported product $\mathbb{D}(f) := \prod_{\beta < \eta} \mathbb{P}_\beta$ where then $\mathbb{D}$ has the countable chain condition.

Proof. Fix an arbitrary $f : \eta \rightarrow 2$. We prove the Lemma using induction over the length η . The limit case is true as we use finite support. Thus assume the assertion of the Lemma is true for η and we want to show it is true for products of length $\eta + 1$. Assume for a contradiction that $\prod_{\alpha < \eta+1} \mathbb{P}_\alpha$ (according to f) does not have the countable chain condition. Hence the tree T_η is not a Suslin tree in $V^{\prod_{\alpha < \eta} \mathbb{P}_\alpha}$, as otherwise both forcings T_η and $Sp(T_\eta)$ would have the countable chain condition. But $\prod_{\alpha < \eta} \mathbb{P}_\alpha * T_\eta = \prod_{\alpha < \eta} \mathbb{P}_\alpha \times T_\eta = T_\eta \times \prod_{\alpha < \eta} \mathbb{P}_\alpha$, and the latter is a forcing with the countable chain condition. Indeed as T_η does not touch the Suslinness of any member of $\vec{T}$ besides T_η , $(T_\alpha : \alpha < \eta)$ is an independent sequence of Suslin trees in V^{T_η} , thus by induction hypothesis, $\prod_{\alpha < \eta} \mathbb{P}_\alpha$ has the countable chain condition in V^{T_η} , so it has the countable chain condition in V . $\square$

In particular this means that $\mathbb{D}(r_U)$ as defined above is a forcing with the countable chain condition which writes the real r_U into a pattern of 0 and 1's on the sequence $\vec{T}^0$ of Suslin trees. We let H_0 be a generic filter for $\mathbb{A}_U$

over W_0 and let H_1 denote a $W_0[H_0]$ -generic filter for $\mathbb{D}(r_U)$. The resulting model $W_1 := W_0[H_0][H_1]$ is the ground model for a second iteration we define later.

Lemma 5.6. *Let W_1 be the universe $W_0[H_0][H_1]$*

Then $W_1 := W_0[H_0][H_1]$ is a ccc extension of W_0 which satisfies:

- $r_U(n) = 1$ if and only if T_n^0 has a branch.
- $r_U(n) = 0$ if and only if T_n^0 is special.

Proof. It suffices to show that $W_0[H_0]$ is a Suslin-tree-preserving extension of W_0 . This is clear as $\mathbb{A}(U)$ is Knaster. □

To summarize we arrived at a situation where the real r_U , which captures all the information about a set of suitable models, is written into the sequence of Suslin trees $\vec{T}^0$. Consequently, any transitive, $\aleph_1$ -sized model M which contains $\vec{C}$, $\vec{T}^0$ and which sees that every tree in $\vec{T}$ is destroyed can compute the real r_U and thus has access to a set of suitable models, which in turn can be used to compute $\vec{T}$ inside M in a correct way. This line of reasoning remains sound in all outer ccc extensions of W_1 as we shall see. Thus the ability of finding $\vec{T}$ in suitable models gives rise to the possibility of using $\vec{T}$ for additional coding arguments over W_1 .

Lemma 5.7. *Let W^* be a ccc extension of W_1 then there is a $\Sigma_1(\vec{C}, \vec{T}^0)$ -formula $\Phi(v)$ such that whenever $x \in 2^\omega$ and $W^* \models \Phi(x)$ then x is the almost disjoint code for a suitable model.*

Proof. The formula $\Phi(x)$, is defined as follows:

$\Phi(x)$ if and only if there is a transitive, $\aleph_1$ -sized $\mathbf{ZF}^-$ model N which contains $\vec{C}$ and $\vec{T}^0$ such that the following holds in N :

- N sees a full pattern on $\vec{T}^0$, i.e. for every $n \in \omega$ and every member T_n^0 , N has either a branch through T_n^0 or a function which specializes T_n^0 .
- The pattern on $\vec{T}^0$ corresponds to the characteristic function of a real r and N thinks that there is a pre-suitable N' such that $x \in N'$ and $(S(x) \cap r)$ is a finite set.

This is a $\Sigma_1(\vec{C}, \vec{T}^0)$ -formula, as it is of the form $\exists N(N \models \dots)$.

We will show that whenever x is a real in W^* satisfying $\Phi(x)$, then x is the almost-disjoint code of some suitable model. Recall the sets U' and U defined earlier. First observe that N has access to U , the collection of reals that code the suitable models U' . This follows from the fact that $\vec{T}^0 \in N$;

hence any pattern on $\vec{T}^0$ recognized by N must be the unique pattern coming from W_1 , which corresponds to the characteristic function of the real r_U .

The assertion “ N' is a pre-suitable model” is given by a $\Delta_1(\vec{C})$ formula in N' , and thus is absolute for the transitive model N . Therefore, if N believes that there exists a pre-suitable N' containing x , then this is indeed true in W^* . Since W^* is a ccc extension of W_1 , and therefore also of W_0 , we already know that W^* can express the statement “ $y \in 2^\omega \cap W_0$ ” as “there exists a pre-suitable P with $y \in P$ ”. Consequently, if N thinks that some pre-suitable N' contains x , then $x \in W_0$. Applying (*) from above, we conclude that $x \in U$, which is exactly what we set out to prove. $\square$

6 Definition of W_2

We shall use the just created W_1 to start a second iteration which will code a fresh sequence of Suslin trees $\vec{S}$ into the W_1 -definable sequence $\vec{T}$ of Suslin trees. The definability of $\vec{T}$ will therefore be passed over to $\vec{S}$. The reason for this seemingly redundant choice, is that we later want to force $\mathbf{MA}_{\omega_1}$, which of course will necessarily create a lot of noise on any definable sequence of independent Suslin trees. This noise is a threat to any intentional coding we aim to build in order to have a universe for the Σ_n^1 -uniformization property.

The addition of the second sequence $\vec{S}$ of independent Suslin trees enables us to later use coding with $\vec{S}$ trees along the usual forcing which produces $\mathbf{MA}_{\omega_1}$ and yet be in full control of the codes we create on the $\vec{S}$ -sequence. This effect can not be reproduced when forcing over just W_1 , this is why we need to pass from W_1 to a more suitable generic extension.

The universe $W_2 = W_1^{\mathbb{P}_0 * \mathbb{P}_1}$ is a two step generic extension of W_1 . The first factor $\mathbb{P}_0$ is just an ω_2 -length product of ordinary Cohen forcing with finite support

$$\mathbb{P}_0 := \prod_{\alpha < \omega_2} \mathbb{C}.$$

As is known due to S. Shelah (see [13], Theorem 28.12, Lemma 28.13 for a proof of this which is due to S. Todorćević) every generically added Cohen real r can be used to define a new Suslin tree S_r . To be more precise, if V is our ground model and

$$(e_\alpha : \alpha < \omega_1)$$

is an ω_1 sequence of functions in V which satisfies

1. for every $\alpha < \omega_1$, e_α is an injection from α to ω ;
2. and for every $\alpha < \beta < \omega_1$, $e_\alpha(\xi) = e_\beta(\xi)$ for all but finitely many $\xi < \alpha$,

then

$$S_r := \{r \circ (e_\alpha \restriction \beta) : \alpha, \beta < \omega_1\}$$

is a Suslin tree, as long as r is a Cohen real over V .

Consequently, if $(c_\alpha : \alpha < \omega_2)$ denotes the sequence of Cohen reals in $W_1^{\mathbb{P}_0}$, the sequence of trees

$$\vec{S} := (S_\alpha : \alpha < \omega_2)$$

(note that here we write S_α where we actually mean S_{c_α}) is an independent sequence of Suslin trees. Indeed, any finite product of S_α 's must be a Suslin tree again, as otherwise, if say S_1 and S_2 are such that $S_1 \times S_2$ is not Suslin, then $\mathbb{C} \times \mathbb{C} \times S_1 \times S_2$ is not a ccc forcing. But rearranging the factors yields that S_2 is a Suslin tree in $W_1^{\mathbb{C} \times S_1 \times \mathbb{C}}$, hence $\mathbb{C} \times S_1 \times \mathbb{C} \times S_2$ has the ccc, which is a contradiction.

Our second forcing $\mathbb{P}_1$ uses our already created independent sequence $\vec{T}$ to code up the trees from $\vec{S}$ with a method completely analogous to our forcing $\mathbb{Q}^1$ which we used in the definition of W_0 . That is if we fix an arbitrary $\alpha < \omega_2$ with the associated tree $S_\alpha \subset \omega_1$, and for $\beta < \omega_1$, we let

$$\mathbb{P}_{1,\alpha}^\beta = \begin{cases} T_{\omega_1\alpha+\beta} & \text{if } S_\alpha(\beta) = 1 \\ \text{SP}(T_{\omega_1\alpha+\beta}) & \text{if } S_\alpha(\beta) = 0 \end{cases}$$

where $T_{\omega_1\alpha+\beta}$ just denotes the forcing which adds an ω_1 -branch through $T_{\omega_1\alpha+\beta}$, and $\text{SP}(T_{\omega_1\alpha+\beta})$ is the forcing which specializes $T_{\omega_1\alpha+\beta}$.

Then

$$\mathbb{P}_{1,\alpha} = \prod_{\beta < \omega_1} \mathbb{P}_{1,\alpha}^\beta$$

using finite support. Note that for every $\alpha < \omega_2$, $\mathbb{P}_{1,\alpha}$ has the ccc.

Finally we let

$$\mathbb{P}^1 := \prod_{\alpha < \omega_2} \mathbb{P}_{1,\alpha}$$

again using finite support. The upshot of these manipulations is, that $\vec{S}$ is now a uniformly definable ω_2 -sequence of independent Suslin trees which has a second crucial feature which is not shared by $\vec{T}$, and which we briefly want to sketch: Let

$$W_2 = W_1^{\mathbb{P}_0 * \mathbb{P}^1}$$

and suppose that $\mathbb{Q}$ is a partial order of size $\aleph_1$ in W_2 . There will be an $\gamma < \omega_2$ such that $\mathbb{Q}$ is already in the intermediate model $W(\prod_{\alpha < \gamma} \mathbb{C}) * (\prod_{\alpha < \gamma} \mathbb{P}_{1,\alpha})$ which is an inner model of W_2 . If we decide to force with $\mathbb{Q}$ over W_2 then the resulting generic extension $W_2^\mathbb{Q}$ can be written as

$$W(\prod_{\alpha < \gamma} \mathbb{C}) * (\prod_{\alpha < \gamma} \mathbb{P}_{1,\alpha}) * (\mathbb{Q}) * (\prod_{\alpha \in [\gamma, \omega_2)} \mathbb{C}) * (\prod_{\alpha \in [\gamma, \omega_2)} \mathbb{P}_{1,\alpha}).$$

As a result the Suslin trees which $\prod_{\alpha \in [\gamma, \omega_2)} \mathbb{C}$ creates are still Suslin in $W_2^{\mathbb{Q}}$. This feature will be very useful when we try to force $\mathbf{MA}_{\omega_1}$ and simultaneously a Δ_4^1 -definable wellorder of the reals and the Σ_n^1 -uniformization property.

Lemma 6.1. *The universe W_2 is a cardinal preserving generic extension of W_0 , whose continuum is $\aleph_2^W$. Moreover there is a $\Sigma_1(\omega_1)$ -formula $\Xi(v_0, v_1, \omega_1)$ such that for every $\gamma < \omega_2$,*

$$W_2 \models \forall \beta < \omega_1 (\Xi(\beta, \gamma, \omega_1) \Leftrightarrow \beta \in S_\gamma)$$

Thus initial segments of the sequence $\vec{S}$ are uniformly $\Sigma_1(\omega_1)$ -definable over W_2 .

Proof. The proof of the definability claim is almost identical to the proof of Lemma 4.4, so we omit it. The rest is just a summary of the discussions above. $\square$

7 Coding forcings to alter a Σ_4^1 -predicate

We work now over W_2 . Our goal is to define a Σ_4^1 -predicate $\sigma(x)$ such that

1. $W_2 \models \neg \exists x \sigma(x)$.
2. For an arbitrary real x , there is a forcing $\text{Code}(x)$ which has size $\aleph_1$ and has the ccc and which has the effect that $W_2^{\text{Code}(x)} \models \sigma(x)$, yet for every real $y \neq x$, $W_2^{\text{Code}(x)} \models \neg \sigma(y)$

We shall define the coding forcing $\text{Code}(x)$ along our desired Σ_4^1 -formula σ . Assume that $x \in 2^\omega$ is an arbitrary real in W_2 . In a first step we fix the first ω -block $\vec{S}^1$ from $\vec{S}$ and define the forcing $\text{Code}(x)$ relative to this first ω -block of trees. This choice should decrease notation as we spare an additional ordinal in the definition of Code . We add however that the coding forcing can be defined in exactly the same way for other ω -blocks of Suslin trees from $\vec{S}$.

We use the same technique as in Lemma 5.5 to write x into $\vec{S}^1$, i.e. we first let the first factor of $\text{Code}(x)$ be the finitely supported product of factors defined as follows:

$$\mathbb{P}_n = \begin{cases} S_n^1 & \text{if } x(n) = 1 \\ Sp(S_n^1) & \text{if } x(n) = 0 \end{cases}$$

To form the second forcing we need to find a “nice” subset of ω_1 which codes all the branches and specializing functions along with some suitable models. Let A be a suitable model which computes $\vec{T}^1$ correctly. We collect the set $\mathcal{J}_{\omega_1}^{M_1}$, the relevant branches and specializing functions through elements of $\vec{T}^0$ which create the pattern which codes up r_U , the relevant reflecting sequences to define the suitable A and write everything into one

set $X \subset \omega_1$. Note that if $L_\zeta[X]$ is a $\mathbf{ZF}^-$ -model which contains $X \subset \omega_1$, we obtain that

$$\begin{aligned} L_\zeta[X] \models n \in x \rightarrow S_n^1 \text{ has an } \omega_1\text{-branch and} \\ n \notin x \rightarrow S_n^1 \text{ has a specializing function} \end{aligned}$$

where we define $\vec{T}^1$ and $\vec{S}^1$ inside $A \in L_\zeta[X]$ and A 's suitability is confirmed via $M_1|_{\omega_1}$, the $M_1|_{\omega_1}$ -definable ladder system $\vec{C}$ and the $M_1|_{\omega_1}$ -definable trees $\vec{T}^0$.

We first note that any transitive, $\aleph_1$ -sized $\mathbf{ZF}^-$ model M which contains X will satisfy

$$\begin{aligned} (M, \in, \mathcal{J}_{\omega_1}^{M_1}) \models & \text{“Decoding } X \text{ yields several objects namely} \\ & \text{a model } m \text{ and } m = \mathcal{J}_{\omega_1}^{M_1} = \bigcup_{\mathcal{J}_\eta^{M_1} \in \mathcal{I}} \mathcal{J}_\eta^{M_1}, \\ & \text{a suitable model } a \text{ which in turn computes trees } \vec{s}^1 \\ & \text{some branches } \vec{b} \text{ and specializing functions } \vec{f} \text{ through } \vec{s}^1, \\ & \text{such that for any } \eta < \omega_2 \text{ such that } L_\eta[a][\vec{b}][\vec{f}] \models \mathbf{ZF}^- \\ & L_\eta[a][\vec{b}][\vec{f}] \models \forall n (n \in x \rightarrow S_n^1 \text{ has an } \omega_1\text{-branch} \\ & \quad n \notin x \rightarrow S_n^1 \text{ has a specializing function}) \end{aligned}$$

In particular, this will be true for a $\mathbf{ZF}^- + \text{“}\aleph_1 \text{ exists”}$ model of the form $(L_\xi[X], \in, \mathcal{J}_{\omega_1}^{M_1})$, $\xi < \aleph_2$ as we can always assume that the decoding function is absolute.

We consider the club of countable structures

$$C := \{\eta < \omega_1 : \exists (M, \in, P) < (L_\xi[X], \in, \mathcal{J}_{\omega_1}^{M_1}) (|M| = \aleph_0 \wedge \eta = \omega_1 \cap M)\}$$

If we assume that $(M, \in, P) \in C$, $M \cap \omega_1 = \eta$, then the model $m \in M$ which satisfies the formula $m = \mathcal{J}_{\omega_1}^{M_1}$ locally in M , will also satisfy $m \in \mathcal{I}$ as seen from the outside, as it is the limit of reals which are elements of $\mathcal{I}$ and taking unions is absolute. So $m = \mathcal{J}_\eta^{M_1} \in \mathcal{I}$.

Consequently if $(N, \in)$ is an arbitrary countable transitive model of $\mathbf{ZFC}^-$ and “ $\aleph_1$ exists”, such that $X \cap \omega_1^N \in N$ and $\omega_1^N \in C$ then also $\mathcal{J}_\eta^{M_1} \in N$ and N will decode out of $X \cap \omega_1^N$ with the help of the ladder system and the almost disjoint family computed from $\mathcal{J}_\eta^{M_1}$ exactly what $(\bar{M}, \in, \mathcal{J}_\eta^{M_1})$ decodes out of $X \cap \omega_1^N$, where the latter is the transitive collapse of $(M, \in, P) < (L_\xi[X], \in, \mathcal{J}_{\omega_1}^{M_1})$. In particular, if we denote the Δ_1 -definable decoding functions with

dec_1, dec_2, dec_3 and dec_4 respectively, then we obtain

$$\begin{aligned}
N \models \exists m_1 \exists \vec{c} \exists \vec{b}, \vec{f} (dec_1(X \cap \omega_1^N) = m_1 \wedge dec_2(X \cap \omega_1^N) = a \\
\wedge dec_3(X \cap \omega_1^N) = \vec{b} \wedge dec_4(X \cap \omega_1^N) = \vec{f} \\
\text{and for any } \mathbf{ZF}^- \text{ model of the form } L_\eta[X \cap \omega_1^N] \text{ we have that} \\
L_\eta[X \cap \omega_1^N] \models \forall n (n \in x \rightarrow S_n^1 \text{ has an } \omega_1\text{-branch} \\
n \notin x \rightarrow S_n^1 \text{ is special})),
\end{aligned}$$

where again the trees S_n^1 are computed inside a with the help of the formula which defines $\vec{S}^1$ in W_2 . Further, as $dec_1(X \cap \omega_1^N) = m_1 = \mathcal{J}_\eta^{M_1}$, we get that (from the outside, not from N 's perspective)

$$dec_1(X \cap \omega_1^N) \in \mathcal{I}.$$

Now let the set $Y \subset \omega_1$ code the pair (C, X) such that the odd entries of Y should code X and if $Y_0 := E(Y)$ where the latter is the set of even entries of Y and $\{c_\alpha : \alpha < \omega_1\}$ is the enumeration of C then

1. $E(Y) \cap \omega$ codes a well-ordering of type c_0 .
2. $E(Y) \cap [\omega, c_0) = \emptyset$.
3. For all β , $E(Y) \cap [c_\beta, c_\beta + \omega)$ codes a well-ordering of type $c_{\beta+1}$.
4. For all β , $E(Y) \cap [c_\beta + \omega, c_{\beta+1}) = \emptyset$.

The set $Y \subset \omega_1$ codes X but has the following additional property:

Let M be an arbitrary countable transitive model of $\mathbf{ZF}^- + \text{"}\aleph_1 \text{ exists"}$ for which there is a $\mathcal{J}_\eta^{M_1} \in \mathcal{I}$ (again from the outside, not from M 's perspective) such that $\omega_1^M = \omega_1^{\mathcal{J}_\eta^{M_1}}$ and $\mathcal{J}_\eta^{M_1} \in M$. Assume that $Y \cap \omega_1^M \in M$ then M can decode out of $Y \cap \omega_1$ with the help of the almost disjoint family of $\mathcal{J}_{\omega_1}^{M_1}$ the following objects:

- a transitive model m which M believes to be of size $\aleph_1$,
- a model a which, M believes, is suitable relative to the ladder system and the almost disjoint family of reals defined in $\mathcal{J}_\eta^{M_1}$ and which computes $\vec{s}^1$ using the same formula which computes the trees in W_2 ,
- a set of branches $\vec{b}$ through $\vec{s}^1$.
- and specializing functions $\vec{f}$ through elements of $\vec{s}^1$ such that for any $\mathbf{ZF}^- + \text{"}\aleph_1 \text{ exists"}$ -model of the form $L_\zeta[a, \vec{b}, \vec{f}]$:

$$\begin{aligned}
L_\zeta[a, \vec{b}, \vec{f}] \models \forall n (n \in x \rightarrow s_n^1 \text{ has an } \omega_1\text{-branch} \\
n \notin x \rightarrow s_n^1 \text{ is special})).
\end{aligned}$$

Moreover m is an M_1 initial segment as seen from the outside, i.e. $m = \mathcal{J}_\eta^{M_1} \in \mathcal{I}$.

In the last step, we use almost disjoint coding forcing relative to the $\mathcal{J}_{\omega_1}^{M_1}$ -definable almost disjoint family of reals to obtain a real r_Y which codes our set $Y \subset \omega_1$. Thus we obtain that the following formula $\psi(x, r_Y)$ holds, where $\psi(x, r_Y)$ is defined as:

Assume M is a countable transitive model of $\mathbf{ZF}^- + \text{“}\aleph_1 \text{ exists”}$, and $r_Y \in M$. Assume that for M there is a $\mathcal{J}_\eta^{M_1} \in \mathcal{I}$ such that $\omega_1^M = \omega_1^{\mathcal{J}_\eta^{M_1}}$ and $\mathcal{J}_\eta^{M_1} \in M$. Assume further that $r_Y \in M$ then M , relative to the almost disjoint family of reals from $\mathcal{J}_\eta^{M_1}$, can decode out of r_Y the following

- a transitive model m of $\mathbf{ZFC}^-$ and “ $\aleph_1$ exists”,
- a model a which, M believes, is suitable, and which computes $\vec{s}^1$,
- a set of branches $\vec{b}$ through $\vec{s}^1$.
- and specializing functions $\vec{f}$ through elements of $\vec{s}^1$ such that for any $\mathbf{ZF}^- + \text{“}\aleph_1 \text{ exists”}$ -model of the form $L_\zeta[a, \vec{b}, \vec{f}]$ (computed in m):

$$L_\zeta[a, \vec{b}, \vec{f}] \models \forall n (n \in x \rightarrow s_n^1 \text{ has an } \omega_1\text{-branch} \\ n \notin x \rightarrow s_n^1 \text{ is special})).$$

Moreover m is an M_1 initial segment as seen from the outside, i.e. $m = \mathcal{J}_\eta^{M_1} \in \mathcal{I}$.

A straightforward calculation shows that the statement $\psi(x, r_Y)$ is of the form $(\Sigma_3^1 \rightarrow \Pi_3^1)$, thus it is a Π_3^1 -formula, and stating the existence of such a real r_Y is Σ_4^1 and results in our desired Σ_4^1 -formula σ :

$$\sigma(x) \equiv \exists r \psi(x, r).$$

The existence of a real r witnessing $\psi(x, r)$ is sufficient to conclude that $L[r]$ contains branches through $\aleph_1$ -many trees from $\vec{S}$.

Lemma 7.1. *Let r be such that $\psi(x, r)$ is true. Then, working inside $L[r]$, there is a suitable A such that A computes $\vec{S}^1$ and there are branches $\vec{b}$ and specializing functions $\vec{f}$ such that*

$$\forall n (n \in x \rightarrow S_n^1 \text{ has an } \omega_1\text{-branch} \\ n \notin x \rightarrow S_n^1 \text{ is special})$$

Proof. We note first that $\psi(x, r)$ must also be true (ignoring its statements involving $\mathcal{I}$) for models of uncountable size where we replace $\mathcal{J}_\eta^{M_1}$, in ψ , with $\mathcal{J}_{\omega_1}^{M_1}$. Indeed, if $(M, \in, \mathcal{J}_{\omega_1}^{M_1})$ would be an uncountable, transitive model containing r and $\mathcal{J}_{\omega_1}^{M_1}$ for which $\psi(x, r)$ is false, then we let $(\bar{N}, \in, \mathcal{J}_\eta^{M_1})$ be the transitive collapse of $N < M$, $r, \mathcal{J}_\eta^{M_1} \in \bar{N}$ and $\bar{N}$ would reject $\psi(x, r)$ as well, even though $\bar{N}$ is of the right form as externally $\mathcal{J}_\eta^{M_1} \in \mathcal{I}$, which gives us a contradiction.

But if $\psi(x, r)$ holds for arbitrarily large models M , it must be true in the universe $L[r]$. Indeed if some $\aleph_1$ -sized $\mathbf{ZF}^-$ -model of the form $L_\zeta[M, A, \vec{B}, \vec{F}]$, where $M, A, \vec{B}, \vec{F}$ are just the unions of the computations of $m, a, \vec{b}$ and $\vec{f}$ in suitable countable transitive models of increasing (with limit ω_1) ordinal height, then first note that $M = M_1 | \omega_1$ and $L_\zeta[M, A, \vec{B}, \vec{F}]$ sees that there are branches $\vec{B}$ and specializing functions $\vec{F}$ such that

$$L_\zeta[M, A, \vec{B}, \vec{F}] \models \forall n (n \in x \rightarrow S_n^1 \text{ has an } \omega_1\text{-branch} \wedge n \notin x \rightarrow S_n^1 \text{ is special}).$$

and $L_\zeta[M, A, \vec{B}, \vec{F}]$'s computation of $\vec{S}^1$ must be correct. As the existence of an ω_1 -branch through S_n^1 , and the existence of a specializing function for S_n^1 is upwards absolute we obtain the assertion of the lemma. $\square$

As an immediate consequence of the preceding lemma, the truth of the Σ_4^1 -formula σ has absolute consequences for the ambient universe: whenever $\sigma(x)$ holds, the characteristic function of x is genuinely written into $\vec{S}^1$. This property ensures that our iteration does not inadvertently introduce spurious codes; that is, no unintended real x will satisfy $\sigma(x)$.

Furthermore, we state again that while we initially defined $\text{Code}(x)$ with respect to $\vec{S}^1$, this choice was arbitrary. We can generalize the definition of $\text{Code}(x)$ to utilize $\vec{S}^\eta$ for any $\eta < \omega_2$.

Definition 7.2. *Let $\eta < \omega_2$ and let x be a real. The coding forcing at level η is defined as:*

$$\text{Code}(x, \eta) := \left(\prod_{n \in x} S_n^\eta \times \prod_{n \notin x} Sp(S_n^\eta) \right) * \mathbb{A}(Y)$$

In this generalized form, $\text{Code}(x, \eta)$ operates exactly as the previously defined coding forcing, but it utilizes the η -th ω -block of $\vec{S}$ instead of $\vec{S}^1$.

Moving forward, we adopt the convention that whenever a coding forcing $\text{Code}(x, \eta)$ is used as a factor in an iteration, η is always chosen to be the least ordinal such that no forcing of the form $\text{Code}(y, \eta)$ has appeared in any prior stage. In other words, the iteration uses the ω -blocks of $\vec{S}$ in consecutive order, leaving no unused gaps.

8 Appropriate Σ_n^1 -predicates

We shall use the Σ_4^1 -predicate “being coded into $\vec{S}$ ” (we will often write just “being coded” for the latter) to form appropriate Σ_n^1 -predicates Φ^n for every $n \in \omega$. These predicates share the following properties:

1. $W_2 \models \forall x \neg(\Phi^n(x))$
2. For every real $x \in W_2$, there is an iteration of coding forcings $\text{Code}^n(x) \in W_2$ such that after forcing with it, $W_2^{\text{Code}^n(x)} \models \Phi^n(x)$, and for every real $y \neq x$, $W_2^{\text{Code}^n(x)} \models \neg\Phi^n(y)$.

Most importantly, these properties remain true even when iterating the (iterations of coding forcings $\text{Code}^n(x_i)$ for a sequence of (names of) reals.

The predicates $\Phi^n(x)$ will be defined now. In the following we let (x, y) denote a real z which recursively codes the pair of reals consisting of x and y . Likewise $(x_0, \dots, x_n)$ is defined.

- $\Phi^4(x, y, m) \equiv \exists a_0((x, y, m, a_0) \text{ is coded into } \vec{S})$.
- $\Phi^5(x, y, m) \equiv \exists a_0 \forall a_1((x, y, m, a_0, a_1) \text{ is not coded into } \vec{S})$.
- $\Phi^6(x, y, m) \equiv \exists a_0 \forall a_1 \exists a_2((x, y, m, a_0, a_1, a_2) \text{ is coded into } \vec{S})$.
- $\Phi^7(x, y, m) \equiv \exists a_0 \forall a_1 \exists a_2 \forall a_3((x, y, m, a_0, a_1, a_2, a_3) \text{ is not coded into } \vec{S})$.
- ...
- ...
- $\Phi^{2n+1}(x, y, m) \equiv \exists a_0 \forall a_1 \dots \forall a_{2n-2}((x, y, m, a_0, \dots, a_{2n-2}) \text{ is not coded into } \vec{S})$.
- $\Phi^{2n+2}(x, y, m) \equiv \exists a_0 \forall a_1 \dots \exists a_{2n-1}((x, y, m, a_0, \dots, a_{2n-1}) \text{ is coded into } \vec{S})$.
- ...
- ...

Each predicate Φ^n is exactly Σ_n^1 . In the choice of our Σ_n^1 -formulas $\Phi^n(x)$, we encounter again a periodicity phenomenon, that is two different cases depending on $n \in \omega$ being even or odd, a theme which is pervasive in this area. It is clear that for each predicate Φ^n and each given real x there is a way to create a universe in which $\Phi^n(x)$ becomes true using our coding forcings. We just need to iterate the relevant coding forcings using countable support. For $n = 3$ we just need one coding forcing, for $n \geq 3$ our iteration

will have inaccessible length in order to catch our tail. As shown already, our coding method allows us to exactly code the tuple of reals we want to code, without accidentally adding some unwanted information. Thus the next lemma is straightforward to prove, so we just state it.

Lemma 8.1. *Let W_2 be the ground model. Let $n \in \omega$ and let x be a real in our ground model. Then there is a forcing $\text{Code}^n(x)$ such that if $G \subset \text{Code}^n(x)$ is generic, $W_2[G]$ will satisfy $\Phi^n(x)$ and for every $y \neq x$, $W_2[G] \models \neg\Phi^n(y)$. This property can be iterated, that is it remains true if we replace W_2 with $W_2[G]$ in the above.*

9 Forcing MA and global Σ -uniformization and a definable wellorder

In the final step, we use W_2 as our ground model, use the coding forcings and the appropriate Σ_n^1 -predicates and force in an ω_2 -length, finitely supported iteration MA_{ω_1} while simultaneously work towards global Σ -uniformization.

9.1 Definition of the final iteration

For the iteration we need a bookkeeping function $F : \omega_2 \rightarrow H(\omega_2)$. Many such functions will work, we demand that for every $x \in H(\omega_2)$, the pre-image $F^{-1}(x)$ is unbounded in ω_2 . In the iteration we will distinguish two sets of stages. One set of stages we will use to produce a model of MA_{ω_1} , the other set of stages we use to work for the global Σ -uniformization property.

In order to prevent the forcings we use to get MA_{ω_1} from damaging our codes we create to work towards global Σ -uniformization, we will force MA_{ω_1} in a “diagonal way” (see also [12]). We define a finitely supported iteration $((\mathbb{P}_\alpha, \dot{Q}_\alpha) : \alpha < \omega_2)$ of ccc forcing over W_2 inductively using our fixed F .

We assume that we are at stage $\beta < \omega_2$ and we have already defined the iteration $\mathbb{P}_\beta$ up to $\beta < \omega_2$. We let G_β denote the generic filter for $\mathbb{P}_\beta$. We also assume by induction that every of the first β -many ω_1 -blocks of trees $\vec{h}^\gamma$, $\gamma < \beta$ have already been used for codings, but the elements of the sequences $(\vec{h}^\eta : \eta \geq \beta)$ still form an independent sequence of Suslin trees in $W_2[G_\beta]$. The forcing we have to use next is determined by the value of $F(\beta)$. We distinguish three cases. The first case will work towards MA_{ω_1} , the second one towards the Δ_4^1 -definable wellorder of the reals and the third one works towards the Σ_m^1 -uniformization for $m \geq 4$.

1. We assume that the bookkeeping function F at β hands us an ω tuple $(\gamma_0, \gamma_1, \dots)$ of ordinals below $\aleph_2$ which corresponds to an ω -tuple of $\mathbb{P}_\beta$ -names $(\dot{x}, \dot{m}, \dot{\alpha}, \dot{b}_0, \dot{b}_1, \dot{b}_2, \dots)$ where $\dot{x}$ and $\dot{b}_n$ are $\mathbb{P}_\beta$ -names of a reals and $\dot{m}$ and $\dot{\alpha}$ are $\mathbb{P}_\beta$ -names of a natural number and of an ordinal bigger than 0 and less than the size of the continuum respectively (in fact at

each stage we will only need finitely many of those names of reals $\dot{b}_i$ and the rest of the information will be discarded). The correspondence is given by demanding that $\gamma_0 \leq \beta$ and $\dot{x}$ is the γ_1 -th name of a real in $W_2[G_{\gamma_0}]$, $\dot{b}_0$ is the γ_2 -th name of a real in $W_2[G_{\gamma_0}]$ and so forth. In this situation we will work towards global Σ -uniformization detailed in the next sections.

2. If $F(\beta) = (\dot{x}_0, \dot{x}_1)$ and $\dot{x}_0, \dot{x}_1$ are both $\mathbb{P}_\beta$ -names of reals and x_0, x_1 are the evaluations of these names using G_β , then we look at the $<_{M_1}$ -least names σ_0, σ_1 such that $\sigma_0^{G_\beta} = x_0, \sigma_1^{G_\beta} = x_1$. We assume without loss of generality that $\sigma_0 < \sigma_1$. Then we force with $\dot{\mathbb{Q}}_\beta^{G_\beta} = \text{Code}((0, x_0, x_1, \eta))$, where $(0, x_0, x_1)$ is a real coding the ordered triple $(0, x_0, x_1)$ and η is the least ordinal $< \omega_2$ such that $\mathbb{P}_\beta$ has no factor of the form $\text{Code}(z, \eta)$, in other words η is the first ordinal which indexes an ω -block of Suslin trees not yet used for coding. The 0 should just indicate that we are coding the wellorder of the real and has no deeper meaning.
3. If $F(\beta) = (\delta, \gamma)$, then we look at the γ -th forcing $\mathbb{B}$ of size $\aleph_1$ in $W_2[G_\delta]$ which has the countable chain condition as seen in the universe $W_2[G_\beta]$. We can consider the iteration $((\mathbb{P}_\zeta : \zeta < \beta) * \mathbb{B})$ which has a dense subforcing of size $\aleph_1$ (the dense set is just the set of conditions in $\mathbb{P}_\beta * \mathbb{B}$ which are fully decided), ccc forcing in W_2 and can thus be seen as a subset of ω_1 in W_2 . As $W_2 = W_1[H_{\omega_2}]$, there is a stage $\nu < \omega_2$ such that (a forcing equivalent to) $((\mathbb{P}_\zeta : \zeta < \beta) * \mathbb{B}) \in W_1[H_\nu]$. Now if $\nu \leq \beta$ we let $\dot{\mathbb{Q}}_\alpha^{G_\alpha}$ be $\mathbb{B}$. Otherwise we force with the trivial forcing.

The definition of the third case ensures that we will not accidentally write an unwanted pattern when forcing for MA_{ω_1} and for the Δ_4^1 -definable wellorder of the reals.

Lemma 9.1. *Assume we are at stage α of our iteration and we are in the nontrivial part of case 2 of the definition, thus we force with an $\aleph_1$ -sized $\dot{\mathbb{S}}_\alpha^{G_\alpha} = \mathbb{B}$. Then, if $G_{\alpha+1}$ is $\mathbb{R}_{\alpha+1} = \mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha$ -generic over $W[H_{\omega_2}]$, all the Suslin trees in $\vec{h}_\zeta$, $\zeta \geq \alpha$ remain Suslin trees in $W[H_{\omega_2}][G_{\alpha+1}]$.*

Proof. Assume that we are at stage α , thus the model we have produced so far is $W_1[H_{\omega_2}][G_\alpha]$ and we force with $\dot{\mathbb{S}}_\alpha$ which is a ccc forcing of size $\aleph_1$ in $W_1[H_\nu][G_\alpha]$ for $\nu \leq \alpha$. Consider some block $\vec{h}^\zeta$, $\zeta \geq \alpha$ in the universe $W_1[H_{\omega_2}][\mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha]$. The latter universe is obtained via the iteration

$$(\mathbb{Q}_\zeta : \zeta < \omega_2) * \mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha = (\mathbb{Q}_\zeta : \zeta \leq \nu) * (\mathbb{Q}_\zeta / \mathbb{Q}_\nu : \zeta > \nu) * \mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha$$

and the right hand side can be rewritten as

$$(\mathbb{Q}_\zeta : \zeta \leq \nu) * ((\mathbb{Q}_\zeta / \mathbb{Q}_\nu : \zeta > \nu) \times (\mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha)).$$

As we can switch the order in products, the latter can be written as

$$(\mathbb{Q}_\zeta : \zeta \leq \nu) * ((\mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha) \times (\mathbb{Q}_\zeta/\mathbb{Q}_\nu : \zeta > \nu)),$$

and consequently the $W_1[H_\nu]$ -generic filter H_{ν, ω_2} for the tail $(\mathbb{Q}_\zeta/\mathbb{Q}_\nu : \zeta > \nu, \zeta < \omega_2)$ remains generic over the model $W_1[(\mathbb{Q}_\zeta : \zeta \leq \nu) * (\mathbb{R}_\alpha * \dot{\mathbb{S}}_\alpha)]$. This means in particular that the generically added Suslin trees $\vec{h}^\zeta, \zeta > \nu$ for Cohen forcing for adding a Suslin tree which are elements in H_{ν, ω_2} remain generic even over the ground model $W_1[H_\nu * G_{\alpha+1}]$. Now Cohen forcing is computed in an absolute way in every universe with the same ω_1 , and trivially, every generic filter for it is a Suslin tree. Hence we obtain that every $h^\zeta_\eta, \zeta > \nu, \eta < \omega_1$ is a Suslin tree in $W_1[H_\nu * G_{\alpha+1}]$, thus every $h^\zeta_\eta, \zeta \geq \alpha \geq \nu$ is a Suslin tree in $W_1[H_{\omega_2} * G_{\alpha+1}]$ as claimed. $\square$

9.2 Proof that $\mathbf{MA}_{\omega_1}$ holds true

We iterate the above defined forcing for ω_2 -many stages and let $W_2[G_{\omega_2}]$ denote the resulting generic extension. First we argue that $\mathbf{MA}_{\omega_1}$ holds true there.

Lemma 9.2. *Let G_{ω_2} be a generic filter for $\mathbb{P}_{\omega_2}$. Then in $W_2[G_{\omega_2}]$ $\mathbf{MA}_{\omega_1}$ holds.*

Proof. Let $\mathbb{B} \in W_2[G_{\omega_2}]$ be an $\aleph_1$ -sized forcing with the countable chain condition. Then there will be $\aleph_2$ -many stages in our iteration where the bookkeeping function F considers $\mathbb{B}$. Moreover we can assume that $\mathbb{B}$ will in fact be an element of the inner model $W_1[H_\nu][G_\nu]$ for some $\nu < \omega_2$. But then the rules of the iteration tell us that we have to force with $\mathbb{B}$ eventually in the iteration, so $\mathbf{MA}_{\omega_1}$ is true. $\square$

The presence of $\mathbf{MA}_{\omega_1}$ in the Σ_3^1 -generically absolute $W_2[G_{\omega_2}]$ yields the desired regularity properties on the third projective level, by an argument of G. Hjorth, [7], Cor. 2.4

Lemma 9.3. *In $W_2[G_{\omega_2}]$, all Σ_3^1 -sets are Lebesgue measurable and have the Baire property.*

Proof. Let A be a Σ_3^1 set of reals, let x be a real and let $\phi(v_0, x)$ be a $\Sigma_3^1(x)$ -formula which defines A . We aim to show that A is Lebesgue measurable. We construct the Martin-Solovay tree T_2 in M_1 and note that $|L[T_2] \cap \omega^\omega| \leq \aleph_1$. Consequently, $L[T_2]$ contains at most $\aleph_1$ many Borel codes for Lebesgue measure zero sets.

Martin's Axiom implies that the real line cannot be covered by $\aleph_1$ many measure zero sets. Therefore, the union of all Borel null sets coded in $L[T_2]$ has measure zero in $W_2[G_{\omega_2}]$. Recall that a real $m \in W_2[G_{\omega_2}]$ is random over $L[T_2]$ if and only if it avoids all measure zero Borel sets coded in $L[T_2]$.

Thus by MA_{ω_1} , Lebesgue almost every real in $W_2[G_{\omega_2}]$ is random over $L[T_2]$. Now, consider any such random real m . Because T_2 projects to the universal Σ_3^1 set and as T_2 is also correctly computed in $W_2[G_{\omega_2}]$ as the forcing has size less than the Woodin cardinal, we have that $L[T_2, m, x] <_{\Sigma_3^1} W_2[G_{\omega_2}]$. Following Solovay's argument for projective measurability, we evaluate the truth of $m \in A$ using the random algebra. Inside $L[T_2, x]$, we can associate a Borel set B representing the Boolean value $\|\phi(\dot{m}, \check{x})\|$ in the random forcing, where $\dot{m}$ is the canonical name for the generic random real.

By the forcing theorem, for any real m that is random over $L[T_2, x]$, $L[T_2, m, x] \models \phi(m, x)$ if and only if $B \Vdash \phi(\dot{m}, x)$ if and only if $m \in B^*$. By the Σ_3^1 absoluteness established above, $W_2[G_{\omega_2}] \models \phi(m, x)$ if and only if $L[T_2, m, x] \models \phi(m, x)$.

Therefore, for almost every real $m \in W_2[G_{\omega_2}]$, $m \in A$ if and only if $m \in B$. This means the symmetric difference $A \Delta B$ is contained within the set of reals that are not random over $L[T_2]$. Since the set of non-random reals has Lebesgue measure zero, A differs from the Borel set B by a null set. Hence, A is Lebesgue measurable. $\square$

9.3 Strategy to obtain global Σ -uniformization

We shall describe the underlying idea to force the global Σ -uniformization. Before we start, we fix some notation which should simplify things a bit. First, whenever we use an iteration using coding forcings $\text{Code}(x, \eta)$ over the universe $W_2[G_\beta]$, and we want to define the β -th factor $\dot{Q}_\beta^{G_\beta}$ of our iteration namely a forcing of the form $\text{Code}(z, \eta)$, we typically will not specify the ordinal η anymore as it will always be the least ordinal η such that

$$W_2[G_\beta] \models \text{"there is no real } z \text{ such that } \text{Code}(z, \check{\eta}) \text{ is a factor of } \mathbb{P}_\beta\text{"}.$$

We will from now on just write $\text{Code}(x)$, where we actually should have written $\text{Code}(x, \eta)$. Second, we will typically code reals z which in fact are recursive codes for tuples of reals $(x, y, m, a_0, a_1, \dots, a_n)$, for which we just write $\text{Code}(x, y, m, a_0, \dots, a_n)$, instead of $\text{Code}((x, y, m, a_0, \dots, a_n))$. Now back to the discussion of the strategy we aim to use in our desired forcing.

The definition of the factors of the iteration will depend on whether the formula φ_m we consider at our current stage is in Σ_{2n}^1 or in Σ_{2n+1}^1 where $n \geq 2$. We start with the case where φ_m appears first on an even level of the projective hierarchy. Assume that

$$\varphi_m \equiv \exists a_0 \forall a_1 \exists a_2 \dots \exists a_{2n-3} \psi(x, y, a_0, a_1, \dots, a_{2n})$$

is a Σ_{2n}^1 -formula (where $\psi(x, y, a_0, \dots, a_{2n-1})$ is a Π_2^1 -formula quantifying over the two remaining bounded variables a_{2n-2} and a_{2n-1}) and x is a real. We want to find a value for the uniformization function for φ_m at the x -section of A_m , the latter being the projective set in the plane defined by φ_m .

To start, we list all triples of reals

$$((x, y^0, a_0^0), (x, y^1, a_0^1), (x, y^2, a_0^2), \dots)$$

according to some fixed well-order $<$. To rule out a degenerate case, we assume that for any real x , the first such triple in our wellorder is always $(x, 0, 0)$, i.e. for all x , $a_0 = y_0 = 0$. We decide that if $\forall a_1 \exists a_2 \dots \psi(x, 0, 0, a_1, \dots)$ is true, then our uniformizing function for φ_m at x should have value 0. Note that $\forall a_1 \exists a_2 \dots \psi(x, 0, 0, a_1, \dots)$ is Π_{2n-1}^1 .

Otherwise we work under the Σ_{2n-1}^1 -assumption $\exists a_1 \forall a_2 \dots \neg \psi(x, 0, 0, a_1, \dots)$ and continue. There will be a $<$ -least triple $(x, y^\alpha, a_0^\alpha)$, $\alpha \geq 1$, for which $\forall a_1 \exists a_2 \dots \exists a_{2n-3} \psi(x, y^\alpha, a_0^\alpha, a_1, a_2, \dots)$ is true, otherwise the x -section would be empty and there is nothing to uniformize.

The goal will be to set up the iteration in such a way that all triples (x, y^β, a_0^β) , $\beta > \alpha$ will satisfy the following formula, which is Π_{2n}^1 in the parameters $(x, y^\beta, a_0^\beta, m)$ as is readily checked:

$$\forall a_1 \exists a_2 \forall a_3 \dots \exists a_{2n-3} ((x, y^\beta, m, a_0^\beta, a_1, a_2, \dots, a_{2n-3}) \text{ is not coded into } \vec{S}).$$

At the same time the definition of the iteration will ensure that for every $\beta \leq \alpha$ our tuple (x, y^β, a_0^β) will satisfy

$$\exists a_1 \forall a_2 \exists a_3 \dots \forall a_{2n-3} ((x, y^\beta, m, a_0^\beta, a_1, a_2, \dots, a_{2n-3}) \text{ is coded into } \vec{S}).$$

Provided we succeed, the pair (x, y^α) will then be the unique solution to the following formula, which is Σ_{2n}^1 , and which shall be the defining formula for our uniformizing function:

$$\begin{aligned} \sigma_{\text{even}}(x, y, m) \equiv & \exists a_0 (\forall a_1 \exists a_2 \dots \psi(x, y, a_0, a_1, \dots) \wedge \\ & \neg (\forall a_1 \exists a_2 \dots ((x, y, m, a_0, a_1, a_2, \dots) \text{ is not coded into } \vec{S}))) \end{aligned}$$

Indeed, for all $\beta > \alpha$, (x, y^β, a_0^β) can not satisfy the second subformula of Ψ whereas for all $\beta < \alpha$ (note here that $\alpha \geq 1$), (x, y^β, a_0^β) can not satisfy the first subformula, as $(x, y^\alpha, a_0^\alpha)$ is the least such triple.

If we assume that φ_m is on an odd level of the projective hierarchy we will define things in a dual way to the even case. Assume that $\varphi_m \equiv \exists a_0 \forall a_1 \exists a_2 \dots \forall a_{2n-2} \psi(x, y, a_0, a_1, \dots, a_{2n-2})$ is a Σ_{2n+1}^1 -formula and x is a real. We want to find a value for the uniformization function for φ_m at the x -section.

Again, we list all triples of reals $((x, y^0, a_0^0), (x, y^1, a_0^1), (x, y^2, a_0^2), \dots)$ according to some fixed well-order $<$ with $a_0 = y_0 = 0$. We assume that $\exists a_1 \forall a_2 \dots \neg \psi(x, 0, 0, a_0, a_1, \dots)$ (otherwise 0 would be the value for the uniformizing function at x for φ_m) so there will be a $<$ -least triple $(x, y^\alpha, a_0^\alpha)$, $\alpha \geq 1$, for which $\forall a_1 \exists a_2 \dots \forall a_{2n-3} \psi(x, y^\alpha, a_0^\alpha, a_1, a_2, \dots)$ is true, otherwise the x -section would be empty and there is nothing to uniformize.

The goal will be to set up the iteration in such a way that all triples (x, y^β, a_0^β) , $\beta > \alpha$ will satisfy the following formula, which is Π_{2n+1}^1 in the parameters $(x, y^\beta, a_0^\beta, m)$ as is readily checked:

$$\forall a_1 \exists a_2 \forall a_3 \dots \exists a_{2n-2} ((x, y^\beta, m, a_0^\beta, a_1, a_2, \dots, a_{2n-2}) \text{ is coded into } \vec{S}).$$

At the same time the definition of the iteration will ensure that for every $\beta \leq \alpha$ (x, y^β, a_0^β) will satisfy

$$\exists a_1 \forall a_2 \exists a_3 \dots \exists a_{2n-2} ((x, y^\alpha, m, a_0^\alpha, a_1, a_2, \dots, a_{2n-2}) \text{ is not coded into } \vec{S}).$$

Provided we succeed, the pair (x, y^α) will then be the unique solution to the following formula, which is Σ_{2n+1}^1 , and which shall be the defining formula for our uniformizing function:

$$\begin{aligned} \sigma_{\text{odd}}(x, y, m) \equiv & \exists a_0 (\forall a_1 \exists a_2 \dots \psi(x, y, a_0, a_1, \dots) \wedge \\ & \neg (\forall a_1 \exists a_2 \dots ((x, y, m, a_0, a_1, a_2, \dots) \text{ is coded into } \vec{S}))) \end{aligned}$$

9.4 Forcing global Σ -uniformization

This section will make precise the strategy outlined in the previous section. We assume that we have defined already the following list of notions:

- We defined already our iteration $\mathbb{P}_\beta \in W_2$ up to stage β .
- We picked a $\mathbb{P}_\beta$ -generic filter G_β for $\mathbb{P}_\beta$ and work, as usual, over $W_2[G_\beta]$.
- In $W_2[G_\beta]$ we picked a family $\{\pi_\alpha \mid \alpha < 2^{\aleph_0}\}$ of bijections of $(2^{\aleph_0})^\alpha$ and $2^{\aleph_0}$. These bijections should cohere with the older families of bijections, as discussed above. We assume without loss of generality that such a family exists, if not then we alter $W_2[G_\beta]$ with a ccc forcing (namely e.g. an iteration of Cohen forcing followed by an iteration of almost disjoint coding forcing relative to some maximal, almost disjoint family of reals) such that such a family exists in the new universe.

We assume that the bookkeeping function F at β hands us an ω tuple $(\gamma_0, \gamma_1, \dots)$ of ordinals below $\aleph_2$ which in turn defines an ω -tuple of $\mathbb{P}_\beta$ -names $(\dot{x}, \dot{m}, \dot{\alpha}, \dot{b}_0, \dot{b}_1, \dot{b}_2, \dots)$ where $\dot{x}$ and $\dot{b}_n$ are $\mathbb{P}_\beta$ -names of a reals and $\dot{m}$ and $\dot{\alpha}$ are $\mathbb{P}_\beta$ -names of a natural number and of an ordinal bigger than 0 and less than the size of the continuum respectively (in fact at each stage we will only need finitely many of those names of reals $\dot{b}_i$ and the rest of the information will be discarded). The correspondence is given by demanding that $\gamma_0 \leq \beta$ and $\dot{x}$ is the γ_1 -th name of a real in $W_2[G_{\gamma_0}]$, $\dot{b}_0$ is the γ_2 -th name of a real in $W_2[G_{\gamma_0}]$ and so forth.

We let $x = \dot{x}^{G_\beta}$, and define b_n, m, α and the G_β -evaluation of our list of names of reals $(x, \dot{y}_0, \dot{a}_0), \dots$ accordingly. The natural number m is the Gödelnumber $\# \varphi$ of a Σ_n^1 -formula $\varphi(x, y)$ in two free variables. We will need the following notions. For an arbitrary real x , list the triples

$$(x, \dot{y}^0, \dot{a}_0^0), (x, \dot{y}^1, \dot{a}_0^1), (x, \dot{y}^2, \dot{a}_0^2), \dots$$

according to our fixed well-order $<$. To be more precise we list all the $\mathbb{P}$ -names of reals which are recursive codes for triples of $\mathbb{P}$ -names of reals, where $\mathbb{P}$ is a κ -sized partial order. The list should have the property that names of longer iterations always appear after the names of the shorter iterations. We also assume that for every x , $\dot{a}_0 = \dot{y}_0 = 0$ to rule out a degenerate case in the following. This has technical advantages as will become clear as we proceed in the argument. The upshot of this is that whenever $(x, a_0 = 0)$ is in our Σ_n^1 -set A_m and the membership is already witnessed by $a_0 = 0$, then we let 0 be the x -value of our uniformizing function of A_m . Thus we can, without loss of generality, always work under the Σ_{n-1}^1 -assumption that 0 does not witness that $(x, 0)$ is an element of A_m . As we aim for a Σ_n^1 -definition of the uniformizing function, this assumption is harmless with regards of complexity.

In $W_2[G_\beta]$, for each ordinal $\alpha < 2^{\aleph_0}$ we fix bijections $\pi_\alpha : (2^{\aleph_0})^\alpha \rightarrow 2^{\aleph_0}$, which exist as $W_2[G_\beta] \models 2^{\aleph_0} = 2^{\aleph_1} = \aleph_2$. These bijections are of course sensitive to the surrounding universe and we assume that, as we iteratively enlarge our universe via a forcing, the bijections extend each other so that they cohere. To be more precise if $\pi_\alpha^\beta : (2^{\aleph_0})^\alpha \rightarrow 2^{\aleph_0}$ is our chosen family of bijections³ in the universe $W_2[G_\beta]$ which arises at the β -th stage of our iteration, and $\pi_\alpha^\gamma : (2^{\aleph_0})^\alpha \rightarrow 2^{\aleph_0}$, $\beta < \gamma$ is the family of bijections we fix at the γ -th stage $W_2[G_\gamma]$, then for every $\alpha < (2^{\aleph_0})^{W_2[G_\beta]}$,

$$\pi_\alpha^\gamma \upharpoonright (2^\omega \cap W_2[G_\beta]) = \pi_\alpha^\beta.$$

Our goal is to define the forcing $\dot{\mathbb{Q}}_\beta$ we want to use at stage β , and to define the notion of $\alpha_\beta + 1$ -based forcing. We consider various cases for our definition.

9.4.1 Case 1, even projective level

In the first case we write $\varphi_m = \exists a_0 \forall a_1 \dots \exists a_{2n-4} \psi(x, y, a_0, \dots, a_{2n-4})$, where ψ is a Π_3^1 -formula (and φ_m is a Σ_{2n}^1 formula for $n \geq 1$) for the formula with Gödel number m which is handed to us by the value $F(\beta)$. We repeat our assumption from above that $F(\beta)$ determines a tuple $b_1, \dots, b_{2n-4}$ of real numbers, x , and an ordinal $0 < \alpha < 2^{\aleph_0}$ with the assigned reals a_0^α and y^α which

³These families of bijections are responsible for the fact that the coding construction presented in this paper and the coding construction which forces the, say, Π_3^1 -uniformization property or the Π_3^1 -reduction property (see [8] and [10]) can not be combined.

stem from our wellordered list of triples $(x, y^0, a_0^0), (x, y^1, a_0^1), \dots (x, y^\alpha, a_0^\alpha), \dots$
⁴ a we fixed in advance.

Recall that we have our fixed bijection $\pi_\alpha : (2^{\aleph_0})^\alpha \rightarrow 2^{\aleph_0}$.

Letting $\pi_\alpha^{-1}(b_k) := (b_k^\eta \mid \eta < \alpha)$ then case 1 is the condition that

$$\neg\psi(x, y^\eta, a_0^\eta, b_1^\eta, b_2^\eta, \dots, b_{2n-4}^\eta) \text{ is true for every } \eta < \alpha.$$

Then we use the coding forcing

$$\text{Code}(\# \psi, x, y^\alpha, a_0^\alpha, b_1, b_2, \dots, b_{2n-4}) =: \dot{\mathbb{Q}}_\beta^{G_\beta}$$

as the β -th factor of our iteration.

9.4.2 Case 2, even projective level

We again let $\varphi_m = \exists a_0 \forall a_1 \dots \exists a_{2n-4} \psi(x, y, a_0, \dots, a_{2n-4})$ and we assume that the bookkeeping F at β hands us a tuple $b_1, \dots, b_{2n-4}$ of real numbers, x , and $0 < \alpha < 2^{\aleph_0}$ with the assigned reals a_0^α and y^α .

Case 2 is the condition that if $\pi_\alpha^{-1}(b_k) := (b_k^\eta \mid \eta < \alpha)$ then

$$\psi(x, y^\eta, a_0^\eta, b_1^\eta, b_2^\eta, \dots, b_{2n-4}^\eta) \text{ is true for an } \eta < \alpha.$$

In this situation we force with the trivial forcing at the β -th stage.

9.4.3 Case 3, odd projective level

This is the dual case to the first one, but this time m is the Gödel number of a formula which belongs to an odd projective level. We write

$$\varphi_m \equiv \exists a_0 \forall a_1 \dots \forall a_{2n-3} \psi(x, y, a_0, \dots, a_{2n-3}),$$

where ψ is a Σ_3^1 -formula and φ_m is a Σ_{2n+1}^1 -formula.

The bookkeeping F at β hands us a tuple $b_1, \dots, b_{2n-3}$ of real numbers, x , and $0 < \alpha < 2^{\aleph_0}$ with the assigned reals a_0^α and y^α .

Recall that we have our fixed bijection $\pi_\alpha : (2^{\aleph_0})^\alpha \rightarrow 2^{\aleph_0}$.

Case 3 is the condition that if $\pi_\alpha^{-1}(b_k) := (b_k^\eta \mid \eta < \alpha)$ then

$$\neg\psi(x, y^\eta, a_0^\eta, b_1^\eta, b_2^\eta, \dots, b_{2n-3}^\eta) \text{ is true for every } \eta < \alpha.$$

We do not use a forcing in this case.

⁴This list should be injective, that is whenever a triple $(x, (\dot{y}_\alpha)^{G_\beta}, (\dot{a}_\alpha)^{G_\beta})$ appears which appeared already earlier in our list we drop it.

9.4.4 Case 4, odd projective level

This is dual to case 2.

We assume that if $\pi_\alpha^{-1}(b_k) := (b_k^\eta \mid \eta < \alpha)$ then

$$\psi(x, y^\eta, a_0^\eta, b_1^\eta, b_2^\eta, \dots, b_{2n-3}^\eta) \text{ is true for an } \eta < \alpha.$$

In this situation we use the coding forcing of the form

$$\text{Code}(\# \varphi_m, x, y^\alpha, a_0^\alpha, b_1, \dots, b_{2n-3})$$

as the β -th forcing $\dot{\mathbb{Q}}_\beta$ in our iteration.

We use finite support for this iteration and consequently the iteration is a ccc forcing over W_2 hence preserves cofinalities.

This ends the definition of our iteration. What is left is to show that the definition works in that it will produce a universe where σ_{even} and σ_{odd} serve as definitions of uniformizing functions.

10 Proof that Σ_n^1 -uniformization holds in the final model

We shall prove the global Σ -uniformization now in $W_2[G_{\omega_2}]$.

Lemma 10.1. *In $W_2[G_{\omega_2}]$ the following holds:*

1. *Whenever φ_m is on an even projective level, say $\varphi_m \in \Sigma_{2n}^1$ and $\varphi_m = \exists a_0 \forall a_1 \dots \exists a_{2n-3} \psi(x, y, a_0, \dots, a_{2n-3})$ for $\psi \in \Pi_3^1$ and $(x, y^\alpha, a_0^\alpha)$ is such that*

$$W_2[G_{\omega_2}] \models \forall a_1 \exists a_2 \dots \forall a_{2n-3} \psi(x, y^\alpha, a_0^\alpha)$$

Then for each $\beta > \alpha$

$$W_2[G_{\omega_2}] \models \forall a_1 \exists a_2 \dots \forall a_{2n-3} ((\# \psi, x, y^\beta, a_0^\beta, a_1, \dots, a_{2n-3}) \text{ is not coded})$$

2. *If $\varphi_m = \exists a_0 \forall a_1 \dots \forall a_{2n-2} \psi(x, y, a_0, \dots, a_{2n-2}) \in \Pi_{2n+1}^1$ is a formula on the odd level, $\psi \in \Sigma_3^1$ and $(x, y^\alpha, a_0^\alpha)$ is such that*

$$W_2[G_{\omega_2}] \models \varphi_m(x, y^\alpha, a_0^\alpha)$$

Then for each $\beta > \alpha$

$$W_2[G_{\omega_2}] \models \forall a_1 \exists a_2 \dots \exists a_{2n-2} ((\# \psi, x, y^\beta, a_0^\beta, a_1, \dots, a_{2n-2}) \text{ is coded})$$

Proof. We will only proof the first item, as the second is shown in the dual way. Let $\beta > \alpha$. Work in $W_2[G_{\omega_2}]$. We know that $\forall a_1 \exists a_2 \dots \exists a_{2n-3} \psi(x, y^\alpha, a_0^\alpha)$ is true. The way we defined our iteration, in particular case 2, implies that

we can use the bijection $\pi_\beta : (2^\omega)^\beta \rightarrow 2^\omega$, to translate between true infinite disjunctions of the form

$$\bigvee_{\eta < \beta} \psi(x, y^\eta, a_0^\eta, b_1^\eta, \dots, b_{2n-3}^\eta)$$

and “ $(\# \psi, x, y^\beta, a_0^\beta, b_1, b_2, \dots, b_{2n-3})$ is not coded into $\vec{S}$ ”, where $b_i = \pi_\beta((b_i^\eta \mid \eta < \beta))$. As $\forall a_1 \exists a_2 \dots \forall a_{2n-3} \psi(x, y^\alpha, a_0^\alpha)$ is true, this witnesses that our assertion $\forall a_1 \exists a_2 \dots \forall a_{2n-3} ((\# \psi, x, y^\beta, a_0^\beta, a_1, \dots, a_{2n-3})$ is not coded into $\vec{S}$) is true in $W_2[G_{\omega_2}]$. $\square$

Lemma 10.2. *Let $\varphi(x, y) = \exists a_0 \forall a_1 \dots \exists a_{2n-4} \psi(x, y, a_0, a_1, \dots, a_{2n-4})$ be a formula on the even projective level Σ_{2n}^1 where $\psi \in \Pi_3^1$.*

1. *If α is least such that*

$$W_2[G_{\omega_2}] \models \forall a_1 \dots \exists a_{2n-4} \psi(x, y^\alpha, a_0^\alpha, \dots, a_{2n-4}),$$

then

$$W_2[G_{\omega_2}] \models \exists a_1 \forall a_2 \dots \forall a_{2n-4} ((\# \varphi, x, y^\alpha, a_0^\alpha, a_1, a_2, \dots, a_{2n-4}) \text{ is coded.})$$

2. *If $\varphi(x, y) = \exists a_0 \forall a_1 \dots \forall a_{2n-3} \psi(x, y, a_0, a_1, \dots, a_{2n-3})$ is an element of Σ_{2n+1}^1 , $\psi \in \Sigma_3^1$ and α is least such that*

$$W_2[G_{\omega_2}] \models \forall a_1 \dots \forall a_{2n-3} \psi(x, y^\alpha, a_0^\alpha, \dots, a_{2n-3}),$$

then

$$W_2[G_{\omega_2}] \models \exists a_1 \dots \forall a_{2n-3} ((\# \varphi, x, y^\alpha, a_0^\alpha, a_1, \dots, a_{2n-3}) \text{ is not coded.})$$

Proof. We shall show only the first item of the lemma, as the second one is proved in the dual way. As α is the first ordinal for which

$$W_2[G_{\omega_2}] \models \forall a_1 \dots \exists a_{2n-4} \psi(x, y^\alpha, a_0^\alpha, \dots, a_{2n-4}),$$

we know that

$$\forall \beta < \alpha, W_2[G_{\omega_2}] \models \exists a_1 \forall a_2, \dots, \forall a_{2n-4} (\neg \psi(x, y^\beta, a_0^\beta, a_1, \dots, a_{2n-4})).$$

In particular, for every $\beta < \alpha$ there is a real a_1^β such that for every a_2^β there is a real a_3^β and so on such that $\neg \psi$ holds. Using the bijection π_α , we can find an $a_1 = \pi_\alpha((a_1^\beta)_{\beta < \alpha})$ which has the property that for any real a_2 there will be a real $a_3 = \pi_\alpha(a_3^\beta)_{\beta < \alpha}$ such that for any real a_4 and so on $\neg \psi$ is true.

But this translates via the case 1 rule of our iteration to the assertion that

$$W_2[G_{\omega_2}] \models \exists a_1 \forall a_2 \dots \forall a_{2n-4} ((\# \varphi, x, y^\alpha, a_0^\alpha, a_1, a_2, \dots, a_{2n-4}) \text{ is coded.})$$

This is what we wanted. $\square$

Lemma 10.3. *In $W_2[G_{\omega_2}]$ the Σ_{n+1}^1 -uniformization property holds true for every $n \geq 1$.*

Proof. Again we will only consider the case for the even projective levels. We know already that Σ_2^1 and Σ_3^1 -uniformization are true. Let $\varphi \equiv \exists a_0 \forall a_1 \dots \exists a_{2n-4} \psi(x, y, a_0, a_1, \dots, a_{2n-4})$ be an arbitrary Σ_{2n}^1 -formula in two free variables where ψ is Π_3^1 . Let x be a real such that there is a real y with $W_2[G_{\omega_2}] \models \varphi(x, y)$. We list all the triples $(x, y^\alpha, a_0^\alpha)$ according to our well-order $<$. Remember that we defined y^0, a_0^0 to be both 0. If a_0^0 witnesses that $\forall a_1 \exists a_2 \dots (\psi(x, 0, 0, a_1, \dots))$ is true, which is a Σ_{2n-1}^1 -formula, then $\varphi(x, 0)$ will be true in $W_2[G_{\omega_2}]$ and is the value of our uniformizing function there.

If $a_0^0 = 0$ does not witness that $\forall a_1 \exists a_2 \dots (\psi(x, 0, 0, a_1, a_2, \dots))$ is true, which is a Π_{2n-1}^1 -formula then let $\alpha > 0$ be least such that

$$W_2[G_{\omega_2}] \models \forall a_1 \dots (\psi(x, y^\alpha, a_0^\alpha, a_1, \dots)).$$

Then by the last Lemma

$$W_2[G_{\omega_2}] \models \exists a_1 \forall a_2 \dots \exists a_{2n-4} ((\# \varphi, x, y^\alpha, a_0^\alpha, \dots) \text{ is coded})$$

holds true. Note that this formula is Σ_{2n}^1 .

Yet, by the penultimate Lemma, for each $\beta > \alpha$

$$W_2[G_{\omega_2}] \models \forall a_1 \exists a_2 \dots \exists a_{2n-4} ((\# \varphi, x, y^\beta, a_0^\beta, a_1, \dots) \text{ is not coded}).$$

So (x, y^α) is the unique pair satisfying the Σ_{2n}^1 -formula

$$\begin{aligned} & \exists a_0 ((\forall a_1 \exists a_2 \dots \forall a_{2n-4} (\psi(x, y, a_0, \dots)) \wedge \\ & (\exists a_1 \forall a_2 \dots \exists a_{2n-4} ((\# \varphi, x, y, a_0, a_1, \dots, a_{2n-4}) \text{ is coded}))) \end{aligned}$$

Indeed, if $\beta > \alpha$, then the triple (x, y^β, a_0^β) can not satisfy the second sub-formula above. And if $\beta < \alpha$, then (x, y^β, a_0^β) will not satisfy

$$\forall a_1 \exists a_2 \dots \forall a_{2n-4} \psi(x, y^\beta, a_0^\beta, a_1, \dots, a_{2n-4})$$

as α was chosen to be least.

So to summarize the Σ_{2n}^1 -formula

$$\begin{aligned} & \forall a_1 \exists a_2 \dots (\psi(x, 0, 0, a_1, a_2, \dots) \vee \\ & (\exists a_1 \forall a_2 \dots (\neg \psi(x, 0, 0, a_1, a_2, \dots)) \wedge \exists a_0 ((\forall a_1 \exists a_2 \dots \exists a_{2n-4} (\psi(x, y, a_0, \dots)) \wedge \\ & (\exists a_1 \forall a_2 \dots \forall a_{2n-4} ((\# \varphi, x, y, a_0, a_1, \dots, a_{2n-4}) \text{ is coded})))))) \end{aligned}$$

defines the uniformizing function for φ □

11 The Σ_3^1 -uniformization in $W_2[G_{\omega_2}]$

There is one final property missing to prove the main theorem for case $n = 1$, namely we still need to show that Σ_3^1 -uniformization is true in $W_2[G_{\omega_2}]$. To show this we need a different argument, relying on generic absoluteness and the good Σ_3^1 -wellorder of M_1 .

Recall that M_1 denotes the minimal proper class inner model with one Woodin cardinal. It can be arranged as a mouse M , and for every level M_η of M , any Σ_1 -elementary substructure $\bar{M}$ of M_η collapses to some $M_{\eta'}$ for some $\eta' \leq \eta$.

We shall make crucial use of the fact that countable initial segments of M_1 (and their relativizations) obey a Π_2^1 definition. Say a “simple x -mouse” is a sound, Π_2^1 -iterable x -premouse projecting to ω , representing an initial segment of $M_1(x)$ (the canonical M_1 construction over the real x). It is a $\Pi_2^1(x, z)$ statement to say “ z codes a simple x -mouse” (and soundness is also only roughly first-order over M , so it is also Π_2^1 to say “ z codes a sound simple x -mouse”). Every sound simple x -mouse in $M_1(x)$ is in fact a proper segment of $M_1(x)$. We make heavy use of the following generic absoluteness property:

Theorem 11.1. *M_1 is two-step Σ_3^1 generically absolute for small forcings.*

Theorem 11.2. *In any set generic extension of M_1 , denoted by $M_1[G]$, Σ_3^1 -uniformization holds. (In particular, it holds in $W_2[G_{\omega_2}]$).*

Proof of theorem. Let G be set generic over M_1 . Given a real x , let $M_1(x)$ denote the minimal transitive proper class containing x with one Woodin cardinal.

Claim 1. *Let x be a real in $M_1[G]$ (so $M_1[x]$ is a sub-generic extension). Then $M_1[x] = M_1(x)$.*

Proof. We have $M_1[x] \subseteq M_1(x)$ because $M_1 \subseteq M_1(x)$ by the minimality of M_1 , and of course because $x \in M_1(x)$. And $M_1(x) \subseteq M_1[x]$ by the minimality of $M_1(x)$ (as $x \in M_1[x]$ and $M_1[x]$ has the required large cardinal structure to compute the canonical M_1 hierarchy over x). $\square$

Claim 2. $M_1(x) <_{\Sigma_3^1} M_1[G]$.

Proof. Suppose that $M_1[G] \models \phi(y)$ where ϕ is Σ_3^1 and $y \in \mathbb{R} \cap M_1(x)$. Now $M_1[G]$ is a forcing extension of $M_1(x)$, by say forcing $\mathbb{P} \in M_1(x)$. Work in $M_1(x)$. Let κ be large enough that

$$x, y, \mathbb{P} \in N = (M_1|_\kappa)(x);$$

and then let $\lambda > \kappa$ be such that $N^\# \in (M_1|_\lambda)(x)$ (which exists since $M_1(x)$ contains a Woodin cardinal and is therefore closed under ordinary sharps).

Let $X < (M_1|_\lambda)(x)$ with $\mathbb{P}, x, y, N, N^\# \in X$ and X countable. Let M be the transitive collapse of X and write $\bar{\mathbb{P}}$ for the transitive collapse image of $\mathbb{P}$, etc. Let g be M -generic for $\bar{\mathbb{P}}$ and such that $M[g] \models \phi(y)$. Write $\phi(y) = \exists z \psi(y, z)$ with ψ being Π_2^1 . Let $z \in M[g]$ be such that $M[g] \models \psi(y, z)$. Then $\psi(y, z)$ is really true, because the collapse image of $N^\#$ is the true $\bar{N}^\#$, and this extends to give the true $(\bar{N}[g])^\#$, and $(\bar{N}[g])^\#$ thinks $\psi(y, z)$ is true, so it really is true. But $z \in M_1(x)$, so $M_1(x) \models \phi(y)$. $\square$

We can now prove that $M_1[G]$ satisfies Σ_3^1 -uniformization. Let $\phi(u, v)$ be a Σ_3^1 formula in two variables. Recall that there is the Π_2^1 -formula asserting that “ M is a sound, simple x -mouse”. This formula can be used to obtain a $\Sigma_3^1(x)$ -good wellorder of $\mathbb{R} \cap M_1(x)$, uniformly in x , and this leads as usual to Σ_3^1 -uniformization inside $M_1(x)$, but then by the Σ_3^1 -elementarity of the second claim, the same uniformization works in $M_1[G]$.

More explicitly, say $\phi(x, y)$ says $\exists z \psi(x, y, z)$ where ψ is Π_2^1 . Working in $M_1[G]$, we can then uniformize

$$\{(x, y) \mid M_1[G] \models \phi(x, y)\}$$

with the formula ϕ' , where $\phi'(x, y)$ says “there is a sound simple x -mouse M and $z \in \mathbb{R}$ such that:

- $y, z \in M$ (x is automatically in M),
- $M \models \psi(x, y, z)$ and $\langle x, y, z \rangle^\# \in M$,
- there is no proper initial segment M_0 of M and y_0, z_0 such that (M_0, y_0, z_0) have the properties above,
- M thinks that (y, z) is the least pair (y_0, z_0) with the properties above (in its canonical ordering).

By the Π_2^1 -definability of “sound simple x -mouse”, ϕ' is Σ_3^1 . And ϕ' uniformizes ϕ . For (i) the minimality of (M, y, z) above (and that all sound simple mice N, N' , have either $N \trianglelefteq N'$ or $N' \trianglelefteq N$) guarantees

$$\phi'(x, y) \wedge \phi'(x, y_0) \implies y = y_0;$$

and (ii)

$$\phi'(x, y) \implies \phi(x, y);$$

as witnessed by the z above, because

$$M \models \psi(x, y, z) \text{ and } \langle x, y, z \rangle^\# \in M;$$

and (iii) if $\phi(x, y_0)$ then by Claim 2, $M_1(x) \models \exists y \phi(x, y)$, so we get some $(y_1, z_1) \in M_1(x)$ such that $\psi(x, y_1, z_1)$, but then working in $M_1(x)$ we can minimize and get the (M, y, z) as above, so $\phi'(x, y)$. $\square$

12 Forcing over M_n

This section outlines the necessary adjustments to lift the construction from M_1 to the canonical inner model with n Woodin cardinals, M_n . While the architectural framework of the iterations—producing W_0 , W_1 , W_2 , and the final ω_2 -length iteration—remains identical, the descriptive complexities of the core components shift uniformly.

The first critical adjustment concerns the characterization of countable initial segments. For M_n , the relevant notion of iterability is Π_{n+1}^1 -iterability. This suffices to capture the countable initial segments of M_n even in our ccc generic extensions.

Fact 12.1. *Let $M_n[G]$ be an ω_1 -preserving forcing extension of M_n . Then in $M_n[G]$ there is a Π_{n+1}^1 -definable set $\mathcal{I}_n$ of premice which are of the form $\mathcal{J}_\eta^{M_n}$ for some $\eta < \omega_1$. $\mathcal{I}_n$ is defined as*

$$\mathcal{I}_n := \{\mathcal{M} \text{ ctbl premouse} : \mathcal{M} \text{ is } \Pi_{n+1}^1\text{-iterable, } \omega\text{-sound and projects to } \omega\},$$

and the set

$$\{\eta < \omega_1 : \exists \mathcal{N} \in \mathcal{I}_n (\mathcal{N} = \mathcal{J}_\eta^{M_n})\}$$

is cofinal in ω_1 .

The higher complexity of $\mathcal{I}_n$ changes the complexity of our coding predicates. In the M_1 construction, decoding a real from the independent Suslin trees relied on the Π_2^1 -definability of $\mathcal{I}_1$, resulting in a Π_3^1 decoding formula $\psi(x, r_Y)$ and a Σ_4^1 predicate $\sigma(x)$ for “being coded into $\vec{S}$ ”. When working over M_n , the decoding formula must assert that the decoded model is an element of $\mathcal{I}_n$. Because membership in $\mathcal{I}_n$ is Π_{n+1}^1 , the corresponding decoding formula $\psi_n(x, r_Y)$ becomes Π_{n+2}^1 . Consequently, the master predicate stating the existence of a valid code, $\sigma^n(x) \equiv \exists r \psi^n(x, r)$, becomes a Σ_{n+3}^1 formula. This ultimately establishes our Δ_{n+3}^1 -definable wellorder of the reals.

The second necessary fact concerns the generic absoluteness of M_n , which allows us to generalize Hjorth’s argument for Lebesgue measurability and the Baire property to the Σ_{n+2}^1 level. This relies on a generalization of the Martin-Solovay result (see [22], Lemma 3.7) and the closure of M_n under the $x \mapsto M_k^\#(x)$ operator for $k < n$.

Fact 12.2. *For every $n \in \omega$, M_n is Σ_{n+2}^1 -generic absolute for forcings of size less than the smallest Woodin cardinal.*

Using this absoluteness, we can replicate the proof of Lemma 8.2. Instead of using T_2 as in the M_1 case, we utilize the generalized tree T_{n+1} defined in Section 2.4, which is weakly homogeneous and projects to the universal Σ_{n+2}^1 set in M_n . The MA_{ω_1} forced in the final extension ensures that the real

line cannot be covered by $\aleph_1$ many measure zero sets coded in $L[T_{n+1}]$. The Σ_{n+2}^1 -generic absoluteness guarantees that evaluations of Σ_{n+2}^1 statements via the random algebra in $L[T_{n+1}, x]$ remain absolute to $W_2[G_{\omega_2}]$, yielding the measurability and Baire property of all Σ_{n+2}^1 sets.

Finally, to force Σ_{n+3+m}^1 -uniformization for every $m \in \omega$, we shift the base of our alternating coding predicates Φ^k . Since our fundamental coding predicate $\sigma^n(x)$ (for which we adopt the slogan “ x is coded into $\vec{S}$ ”) is now Σ_{n+3}^1 , we define the sequence of properties starting at this level:

- $\Phi^{n+3} \equiv \exists a_0((x, y, m, a_0) \text{ is coded into } \vec{S})$.
- ...
- ...
- $\Phi^{n+k} \equiv \exists a_0 \forall a_1 \dots ((x, y, m, a_0, a_1 \dots) \text{ is coded into } \vec{S})$

The diagonal iteration strategy described in Section 8 remains identical, and aims at decoupling the uniformization coding from the MA_{ω_1} and wellordering requirements. This establishes the main result:

Theorem 12.3. *For any $n \in \omega$, if the canonical inner model with n Woodin cardinals exists, there is a generic extension in which MA_{ω_1} holds, all Σ_{n+2}^1 -sets are Lebesgue measurable and have the Baire property, there is a Δ_{n+3}^1 -definable wellorder of the reals, and Σ_{n+3+m}^1 -uniformization holds for all $m \in \omega$.*

13 Open Questions

We end with a couple of natural questions whose answers would need new ideas.

Question 2. *Is there a universe with global Σ -uniformization and where every projective set is Lebesgue measurable and has the Baire property?*

Question 3. *Is there a universe with global Σ -uniformization and where every projective set is Lebesgue measurable?*

Question 4. *Given an inaccessible, is there a universe where each projective set is Lebesgue measurable and projective separation fails? The same question can be asked with the Baire property replacing the notion of Lebesgue measurability.*

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